14/11/2025

Module 1

Data Warehousing Fundamentals

✓ 1) Describe in details Data Warehouse; also explains its features, Needs, Application, Benefits and Approaches. UTI

✓ 2) Explain in detail Data Warehouse Architecture. UTI

✓ 3) Why is entity-relationship modeling technique not suitable for the data warehouse? How is dimensional modeling different? UTI

✓ 4) Difference between: OLTP & OLAP operations UTI, MU, May23, D23,

✓ 5) Consider a data warehouse for hospital where there are 3 dimensions: UTI
Doctor, Patient & time. Consider 2 measures; "count" and "charge" where charge is the fee that the doctor charges a patient for a visit. For the above eg. create a cube & illustrate the following OLAP operations.
i) Rollup ii) Drill down iii) Slice iv) Dice v) Pivot.

✓ 6) Compare Star Schema & Snowflake Schema UTI, May23

✓ 7) For a supermarket chain the, Consider the foll. dimensions namely Product, store, Time & Promotion. The Scheme contains a central fact table for sales with 3 measures unit-sales, dollar-sales, dollar-cost. UTI, Nov22, Dec24
i) Draw Star Schema.
ii) Calculate the Maximum no. of base facts record for Warehouse with foll. values given below
a) Time periods - 5 years
b) Store - 300 stores reporting daily sales.
c) Product - 40000 products in each store (above 4000 sell in each store daily).
d) Promotion - a sold item may be in only one promotion in a store on a given day.

8) Illustrate major steps in ETL process MU/A, May24, May25
9) Create a cube & describe all 5 OLAP operations.
Consider the quaterly sales of four companies C1, C2, C3, C4. The dimensions are: MU/A, NOV22
a) Time b) Shopping category (men's, Women's, Electronics, Home)
c) Company.
10) Design a Star Schema & Snowflake Schema. MU/A
11) Shortnote: Techniques of data loading D23
12) The college wants to record the marks for the courses completed by students using the dimensions: D23, May25
I) Course II) Student III) Time & a measure Aggregate marks
Create a cube and describe following OLAP operations:
I) Slice II) Dice III) Rollup IV) Drilldown V) Pivot.
13) A social media plantform want to analyze user engagement data to improve content recommendation & user experience. The INTERACTIONS fact table contains information about user interactions, including interactions details, user information, content details and time periods. The dimension tables provide additional context about users, contains, categories and time periods.
Design a star schema and snowflake schema for the same May24
14) Differentiate: ER modeling vs Dimensional Modeling May24 NOV
15) What are the basic building blocks of Data warehouse? May23, may25
16) Design Star Schema for Company sales with 3 dimensions such as Location, Item & Time. May23
17) Differentiate betn Top-down & Bottom-up approaches for building data warehouse. Discuss the merits & limitations of each approach. Also explain the practical approach for designing a data warehouse May23

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* Data Warehouse :-

- The term Data Warehouse was defined by Bill Inmon in 1990, in the following way: "A data Warehouse is a 'Subject-Oriented', 'Integrated', 'Time Variant' and 'Nonvolatile' collection of data in support of management's decision making process."

- The 4 keywords 'subject-oriented', 'Integrated', 'Time Variant' and non-volatile' distinguish data warehouses from other data repository systems, such as relational database system, transaction processing systems, and file systems.

- An Data Warehouse (DW) is a centralized, integrated repository of Historical and current data, collection from multiple sources, organized to support decision-making, ~~repository~~ reporting & data analysis.

• Features of Data Warehouse :

1. Subject Oriented :
Data gives information about a particular subject Instead of about a company's ongoing operations.

2. Integrated :
Data that is gathered into the Data Warehouse from a variety of sources and merged into a coherent whole.

3. Time Variant :
Stores historical data; All the data in the Data Warehouse is identified with a particular time period.

4. Non-Volatile :
Data is stable in a data warehouse. More data is added but data is never removed. This enables management to gain a consistent picture of the business.

• Needs of Data Warehouse :

1. Business User :
Business users require a data warehouse to view summarized data from the past.

2. Store historical data :
Data warehouse is required to store the time variable data from the past.

3. Make strategic decisions :
Some strategies may be depending upon the data in the data warehouse.

4. For data consistency and quality :
To bring uniformity and consistency in data

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- Appln of Data Warehouse:

1) Banking: Widely used in the banking sector to manage the resources available on desk.

2) Healthcare: Patient care analysis & disease trend tracking.

3) Telecommunication: for product promotions, sales decision.

4) Investment & Insurance Sector: DW has primarily used to analyze data patterns, customer trends.

5) Airline: It is used for operations purpose like crew assignment, etc.

- Benefits of Data Warehouse:

1) Delivers enhanced business intelligence:
can effortlessly be applied too business's processes, for instance, market segmentation, sales, etc.

2) Saves time:
There is no need to retrieve the data from multiple sources as users can access data from one place.

3) Enhances data quality and consistency:
Since the data from sources across the organisn is standardized, each dept. will produce results that are consistent.

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4) Improves decision making process:
provides better insights to decision makers

- Approaches of Data Warehouse

1) Top-down Approach
2) Bottom-Up Approach
3) Hybrid Approach.
4) Federated Approach
5) Practical Approach.

* Difference between

	Top down Approach	Bottom-Up Approach
Design	Starts with overall design	Starts with individual components.
Develop time	Longer	faster
Cost	Higher upfront cost	Lower upfront cost.
flexibility	Less flexible	More flexible
Data Quality	Higher	Varies
User involvement	Less	More
Scalability	More	Less
Risk	Higher	Lower

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- Advantage of Top down approach:

1) centralized control and rules.
2) Inherently designed - not a mash-up of disparate data marts.
3) Results are obtained quickly.
4) Represents a data view from the perspective of the enterprise

- Disadvantages of Topdown approach:

1) Building takes longer.
2) High failure risk
3) Expenses are high.

- Advantages of Bottom-up Approach:

1) Risk of failure is Low.
2) This model strikes a good balanced between centralized & localized flexibility.
3) Allows one to create important data mart first.

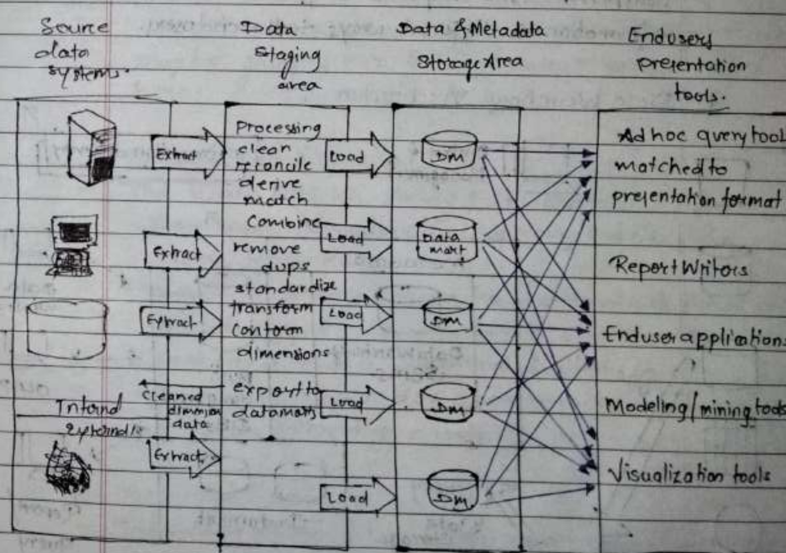
- Disadvantage of Bottom-up.

1) every dataMart is flooded with redundant information
2) Grows unmanageable interfaces.

* A practical Approach:—

A practical approach for designing a DW involves a phased methodology focusing on business needs, iterative development, and flexibility. key steps include define business requirements, assessing data sources, designing the data model and ETL processes and implementing, testing and deploying the Warehouse.

* Basic building blocks of Data Warehouse / Info flow Mechanism:-



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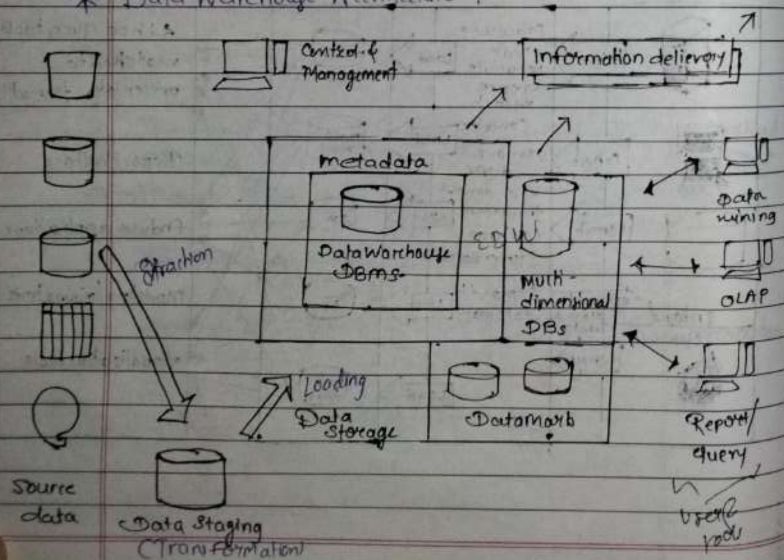
- In fig. the source data component is shown on left, shows various internal and external data source.
- The Next building block is the data staging area where the data is extracted from source data system and then transformation and loading of data is done.
- In middle, data storage component is placed ~~with~~ which manages the data warehouse data.
- A meta data respository is used to keep a track of the data.
- On the Right most side there is information delievery component. This component is responsible to deliever the information in different ways to the end users.

* Data Warehouse Architecture :-



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- Data Warehouse where data is Centralized and stored in one place with organized manner.
- for eg., business stored data, customer data, etc.
- Data Warehouse 4 types of data

① Production ② Archived ③ Internal ④ External

I) Source data :-

Source data coming into the Data Warehouse may be grouped into 4 based categories, as discussed here :

1. Production Data :
- The Data which is coming from Operational System where many transactions are happening and the data coming from it.
- for eg., in a bank, many transaction take place at a sametime and many transactions customer (many) do that type of data is Operational System.
- Working on day-to day operations of any business
- eg., bank, e-commerce web, etc.

2. Archived Data :
- The data of operation system can get discarded or loose by any situation at that time we can use backup data & that type of data is called Archived Data.

3. Internal Data:
Employees in company preserve their "private" spreadsheets, documents and even departmental database. This is Internal Data that could be used in a datawarehouse in some form.

4. External Data:
In this type, the data which is not produced by company but getting from some external source (third-party organizations) such data is called as External Data.

II] Data Staging:-
- Data staging is the place where data is stored on temporal basis.
- The extracted data coming from data source are stored temporarily in a particular form.
- A staging area is where the three major functions of Extraction, Transformation and Loading take place and prepare it for future analysis (ETL process)

• Loading:
- There are 2 types of loading:-
i) Initial loading ii) Incremental loading
- The data which is stored in staging starts loading.

i) Initial loading: Data loaded initially just after company's 1st production.

ii) Incremental loading: Data which is loaded periodically after every next production of company (enterprises).

III] Data Storage:-
- Organized Data with the help of Multidimensional Data Model
- It have's meta Data → which piece of data will go in DataWarehouse.
- Loading is happening because of meta data
कोणसा piece of data DW के अंदर कहा पे जायेगा?
C&M Loading process metadata के acc. हो रहा है क्या?

IV] Data Marts:-
• From EDW the data is divided into marts for various types of data.
• This separated in Marts because in EDW there are many data stored in it
- Data is separated from EDW as per the type of data req'd and store in separate parts in Data marts accordingly.
and because of which fetching of data is easy now.

VI] Information Delivery:-
The Data from DataMart is converted in particular form as users demands and then It is delievered.

VII] Users & tools:-
While delievering the information data in particular format, ~~and it~~ there are various process being applied on it and these tools are there for non-technical background people.

VIII] Control & Management:-
This component of DW Arch. sits on the top of all other components.
It monitors the whole process and interactions going in DataWarehouse Arch.

* Types of MetaData:-

- Operational Metadata
  - ↓ data & its source of location, all info abt Operⁿ D.S. can be back to source datasets after delievery.
- Extraction & transformation Metadata
  - ↓ info abt these
- Enduser Metadata
  - ↓ Ramukaka of library "Navigational map of DW"

* ETL Process (Extraction - Transformation, Loading):-

- The major steps of the ETL process are Extracting data from various sources, Transforming it to be clean and standardized and loading the processed data into a target system like a D.W.
- These 3 core phases work together to prepare raw data for analysis and reporting.

Extract:
This is the first stage, where raw data is collected from one or diverse source systems such as databases, applications, flat files or cloud services. and is exported to a temporary Staging area.

Transform:
This is the core processing phase, where the extracted data is cleaned, converted and standardized.
This involves several sub-process, including:

- Cleansing : Removing errors, filling missing values, correcting inconsistencies.
- Standardization : Ensuring data is in a consistent format across all sources.
- Aggregation : Combining data points to create summarized information
- Deduplication : Removing duplicate records.
- Validation : Verifying the accuracy & integrity of data

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Load:

The final stage where the transformed data is moved into the final target destination.

- A full Load : which overwrites any existing data.
- An incremental load : which adds new or changed data to the existing repository.

* Why is Entity-relationship Modelling technique not suitable for the data warehouse? How is dimensional modelling different?

→ Entity-relationship modelling (ER) is not suitable for data warehouses because:
- focus on transactional data and operational systems.
- Complex and detailed, making it difficult to analyze large datasets.
- Not optimized for quering and reporting.

Dimensional Modeling is different because:
- Focuses on business processes and analytical queries.
- Simplifies complex data into facts and dimensions.
- Optimized for quering, reporting and analysis.
- Uses star or Snowflake Schema for data organization.
- Dimensional Modelling is more suitable for data Warehouse as it Supports business intelligence and decision-making.

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* Difference between:- ER Moddling and Dimensional Modelling

ER (Entity-Relationship) Modelling	Dimensional Modelling
1) ER Modelling have logical & physical model.	1) Dimensional Modelling have physical model.
2) ER modelling is used to design transactional dB for ~~data bases~~ day-to-day operations.	2) Dimensional Modelling is used to design data Warehouses for Analytical processing.
3) Uses Current Data	3) Uses Historical Data
4) It is not mapped for creating Schemas	4) It is mapped for creating Schemas.
5) ER Modelling is used for normalizing the OLTP design.	5) Dimensional Modelling is used for denormalizing the OLAP design.
6) Data Storage is Volatile	6) Data Storage is non-volatile.
7) This structure supports CRUD Operations. (Create, Read, Update, Delete).	7) The Structure supports reporting and analytical queries like SUM, AVG, COUNT, SELECT.
8) It emphasizes data Integrity & Relationships.	8) It emphasizes Query efficiency and Summarization.
9) ER Models are Complex	9) Dimensional Models are Simple.

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* Difference between: Star Schema & Snow Flake Schema

Parameter	Star Schema	Snow-flake Schema
Ease of maintenance /Change	It has redundant data & less easy to maintain /change	It has No Redundancy and more easy to maintain/ change.
Normalization	It has denormalized tables.	It has Normalized tables.
Structure	Star Schema has a central fact table connected directly to 1 dimensional tables. [1 Fact table & 1 or more DT] single dimension table for each dimension	Snowflake Schema has a central fact table connected to Normalized many dimensional table (More than 1 dimension table for each dimension)
Query performance	Queries run Faster.	Queries run slower
Joins	Few Joins	Higher Joins
Ease of use	Simpler to design & understand.	More complex to design & understand.
Space	Uses More storage space	Uses less storage space.
Parent table	No parent table for dimension table.	1 or more Parent table for Dimension table.
When to use	Best for small/medium Warehouse.	Better for Large Warehouse

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* Difference between: OLTP & OLAP

Parameter	OLTP	OLAP
Name	OLTP stands for Online Transaction Processing.	OLAP stands for Online Analytical Processing.
Purpose	OLTP is used to run real time / business processing data (day to day).	OLAP is used to run Past time data, analysis of business.
Response	In Millisecond	In seconds to minutes.
Design	DB design is Application Oriented.	DB design is Subject Oriented.
Operations	Performs INSERT, UPDATE, DELETE, operations.	Perform Rollup, drill-down, Slice, dice & pivot opt.
Handles	OLTP handles many small transactions at a time.	OLAP handles fewer but Complex queries.
Storing	OLTP stores Current & detailed data.	OLAP stores Historical & Summarized data.
Query	OLTP provides fast query, processing for routine tasks.	OLAP provides Slow query, but deep analysis of data.
Example	Eg. ATM withdrawl, railway booking, etc.	Eg. Sales trend analysis, profit analysis, etc.

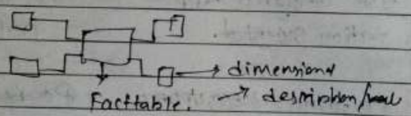
#primary key= uniquely identifies
#Foreignkey = refers to "PK"

Schema: An logical description of an entire Database.
Includes Records of all Record types.

Starschema — Snowflake Schema — fact constellation Schema

① Starschema:

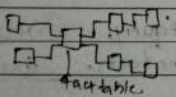
- elementary format Dimensional Model, in which data is organized into facts & Dimensions.



- 1.Facttable
- ~~1ormore~~ Dimension Tables has only 1 dimension

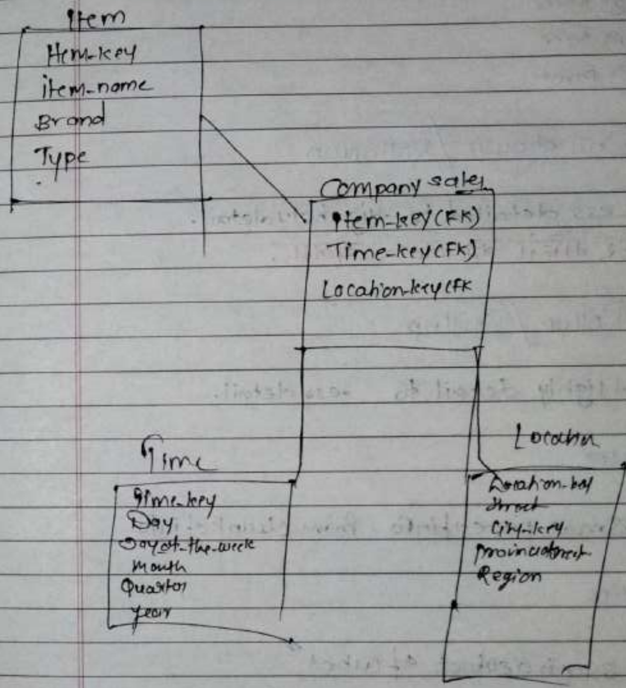
② Snowflake Schema:

- Variant of star schema
- Centralized fact-table is connected to multiple dimensions.



- multiple dimensional tables for each dimensions.

* Design a Star Schema for company sales with 3 dimensions such as Location, Item & time.



have flag 0 in it.

1.7.5 Fact Constellation Schema M25

- Fact Constellation is a schema for representing multidimensional model.
- It is a collection of multiple fact tables having some common dimension tables.
- It can be viewed as a collection of several star schemas and hence, also known as *Galaxy schema*.
- It is one of the widely used schema for Data warehouse designing and it is much more complex than star and snowflake schema.
- For complex systems, we require fact constellations.

1.7.5(A) Advantage of Fact Constellation Schema

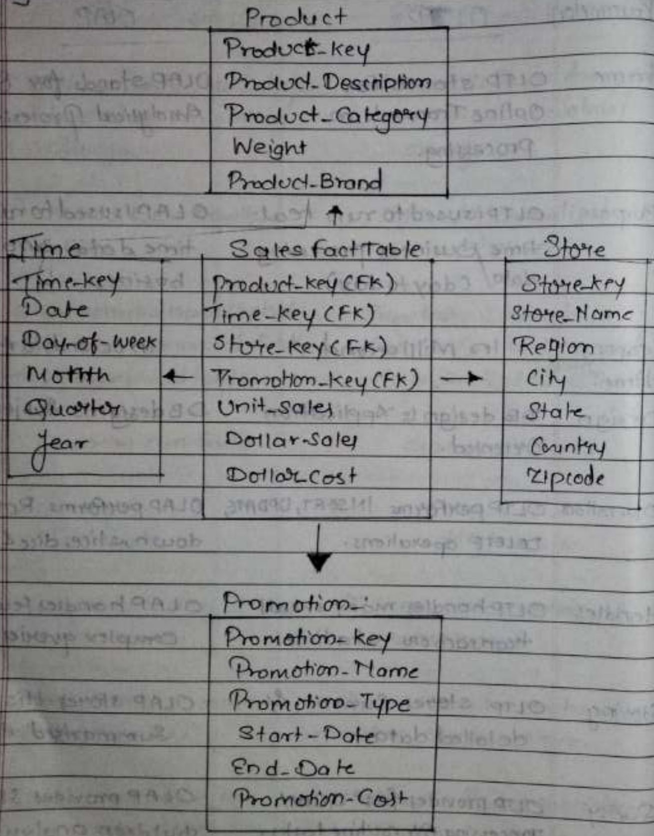
Provides a flexible schema.

1.7.5(B) Disadvantage of Fact Constellation Schema

It is much more complex and hence, hard to implement and maintain.

Q] For a supermarket chain ....

→ i] Star Schema



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ii] Fact Table records

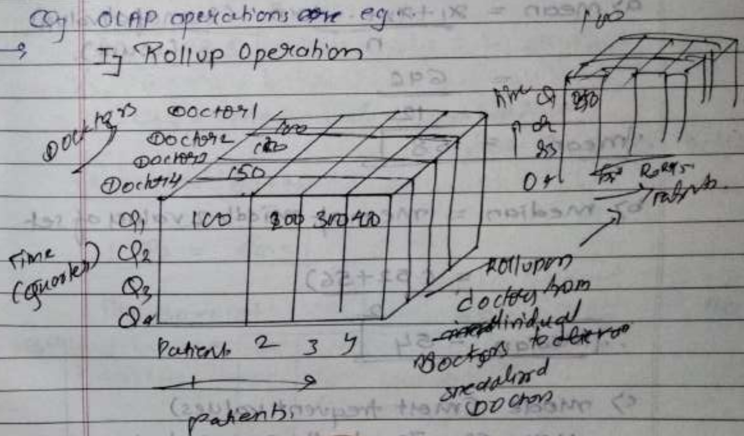
$365 \times 5 = 1825$

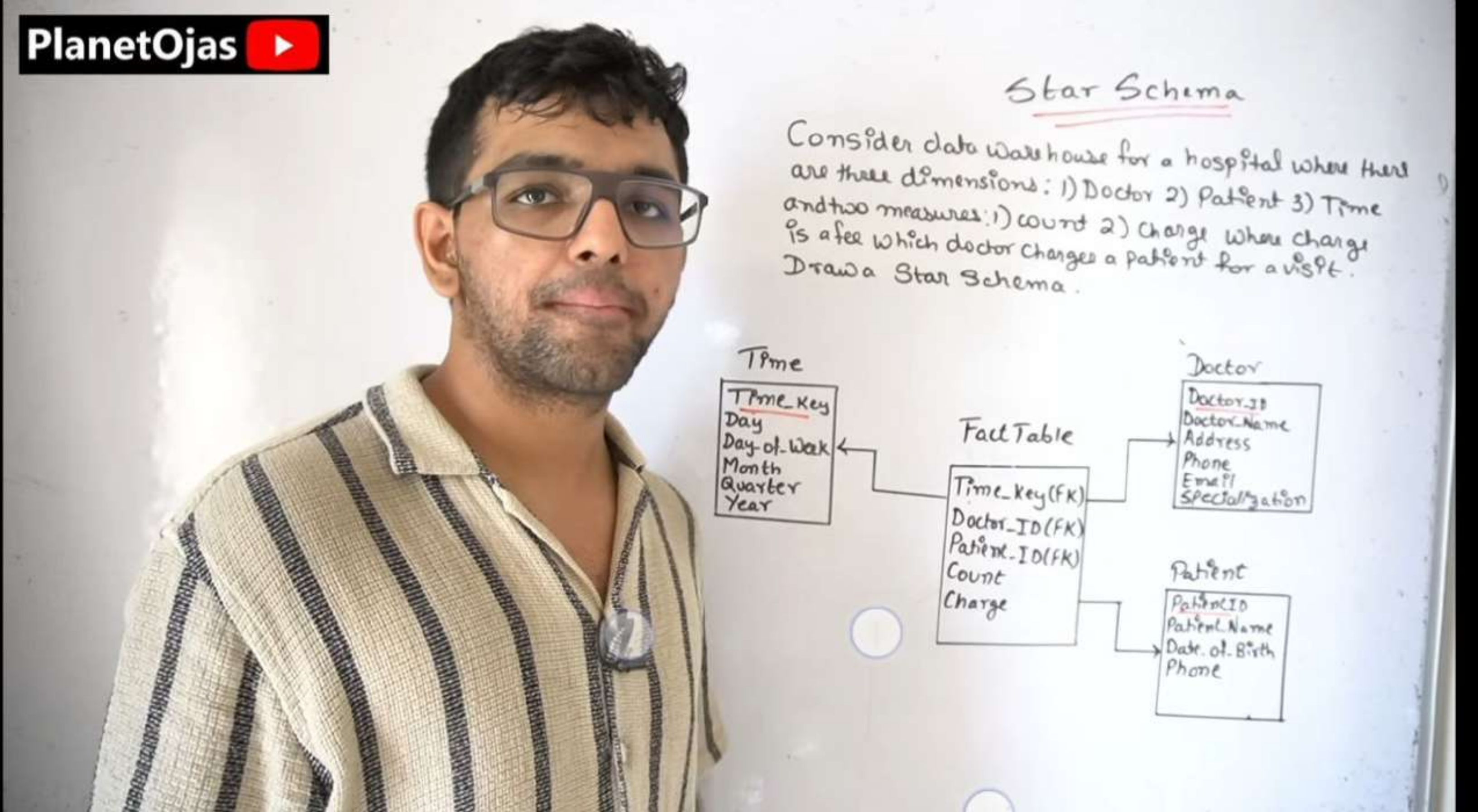
- Time periods = 5 years = 5×365 days = 1825 days
- No. of Stores = 300 stores
- Product in each store dially sell = 4000
- Promotion = 1

∴ Maximum no. of fact table records
= 1825 × 300 × 4000 × 1
= 2190000000 records

Q] OLAP operations are eg.

→ I] Rollup Operation

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Star Schema
Consider data warehouse for a hospital where there are three dimensions: 1) Doctor 2) Patient 3) Time and two measures: 1) count 2) charge where charge is a fee which doctor charges a patient for a visit. Draw a Star Schema.
Time
Time_Key
Day
Day-of-Week
Month
Quarter
Year
Fact Table
Time_Key(FK)
Doctor_ID(FK)
Patient_ID(FK)
Count
Charge
Doctor
Doctor_ID
Doctor_Name
Address
Phone
Email
Specialization
Patient
PatientID
Patient_Name
Date_of_Birth
Phone

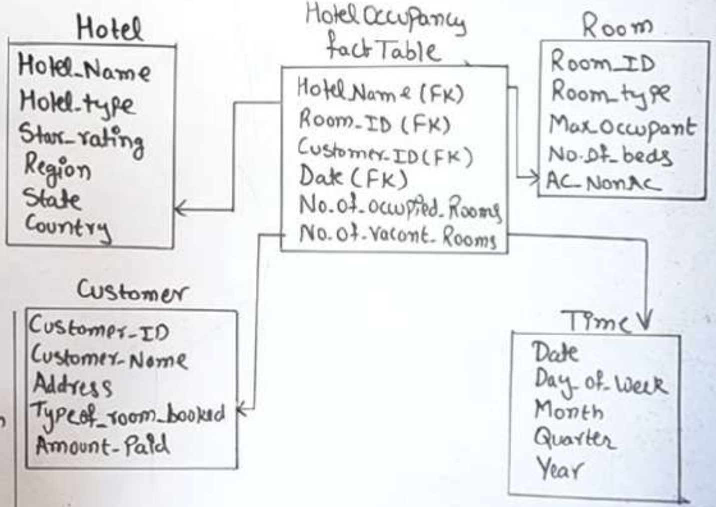
Snow Flake Schema

Customer ID	Customer Name	Location_ID	City	State
A11	Madan Laal	11	Mumbai	Maharashtra
A12	Dhinchak Pooja	12	Agra	Uttar Pradesh
A13	Mandeep Maheshwari	11	Mumbai	Maharashtra
A14	Billey Butcher	11	Mumbai	Maharashtra

⇒ Consider a data Warehouse for hotel occupancy, Where there are four dimensions which are: 1) Hotel 2) Room 3) Time 4) Customer & Two Measures a) Occupied rooms b) Vacant rooms. Draw Snow Flake schema

Customer ID	Customer Name	Location ID
A11	Madan Laal	11
A12	Dhinchak Pooja	12
A13	Mandeep Maheshwari	11
A14	Billey Butcher	11

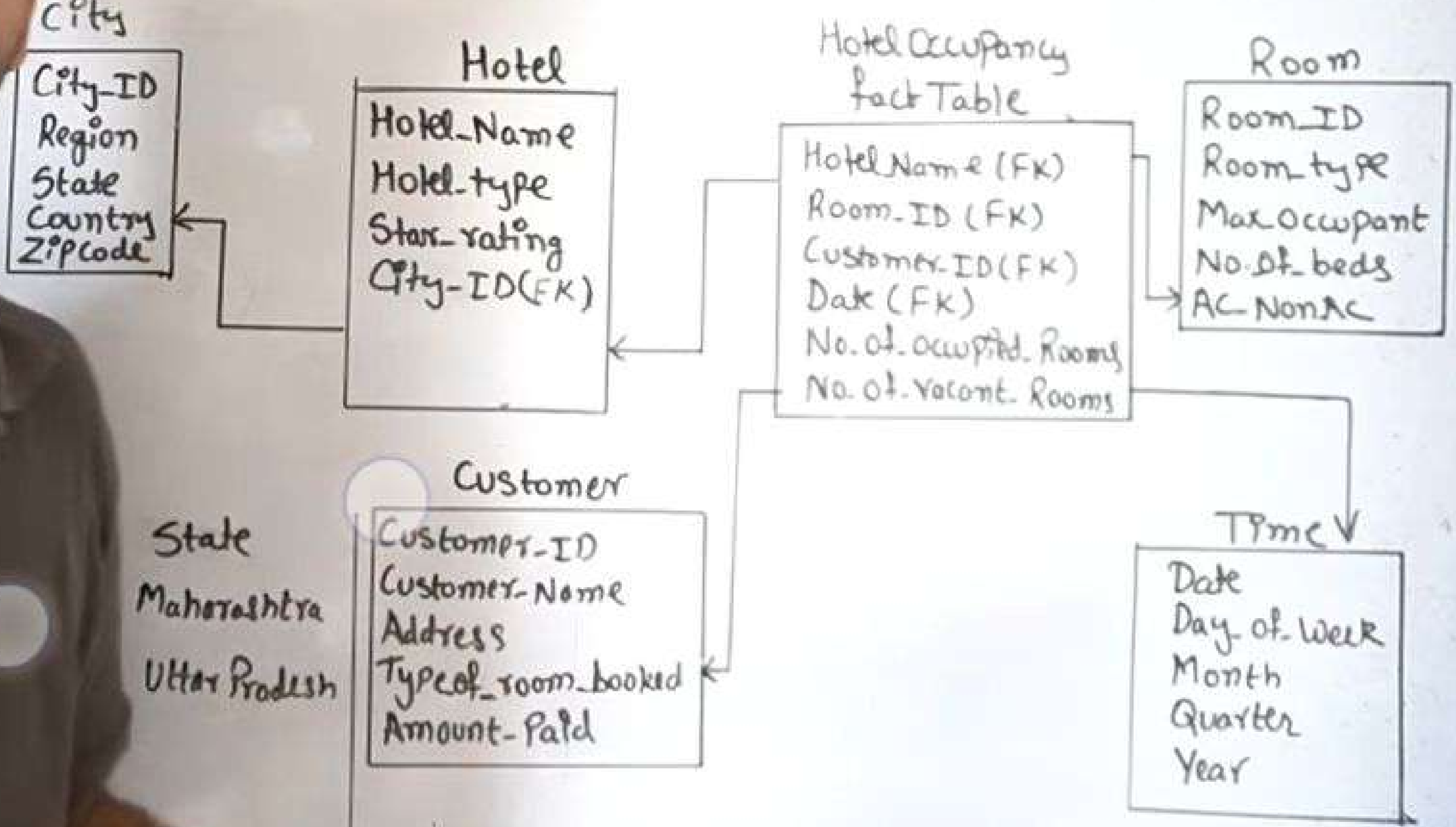
Location ID	City	State
11	Mumbai	Maharashtra
12	Agra	Uttar Pradesh



Snow Flake Schema

te
harashtra
ar Pradesh
arashtra
arashtra

⇒ Consider a data Warehouse for hotel occupancy, Where there are four dimensions which are:
1) Hotel 2) Room 3) Time 4) Customer &
Two Measures a) Occupied rooms b) Vacant rooms
Draw Snow Flake schema



State
Maharashtra
Uttar Pradesh

(a) Star Schema

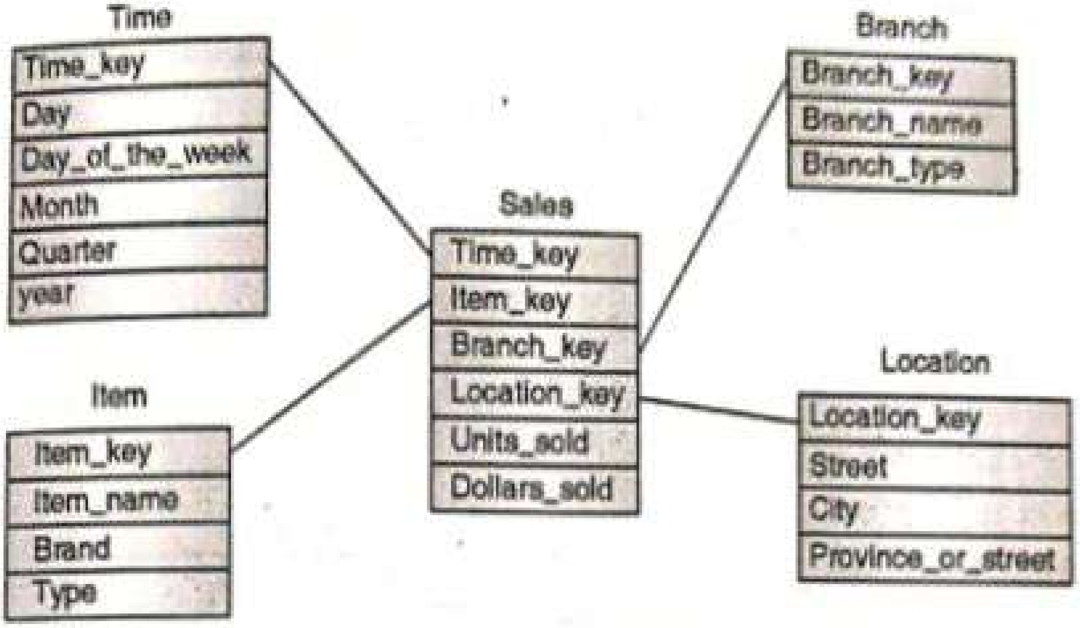


Fig. P. 1.17.1 : Sales Star Schema

(b) Snowflake Schema

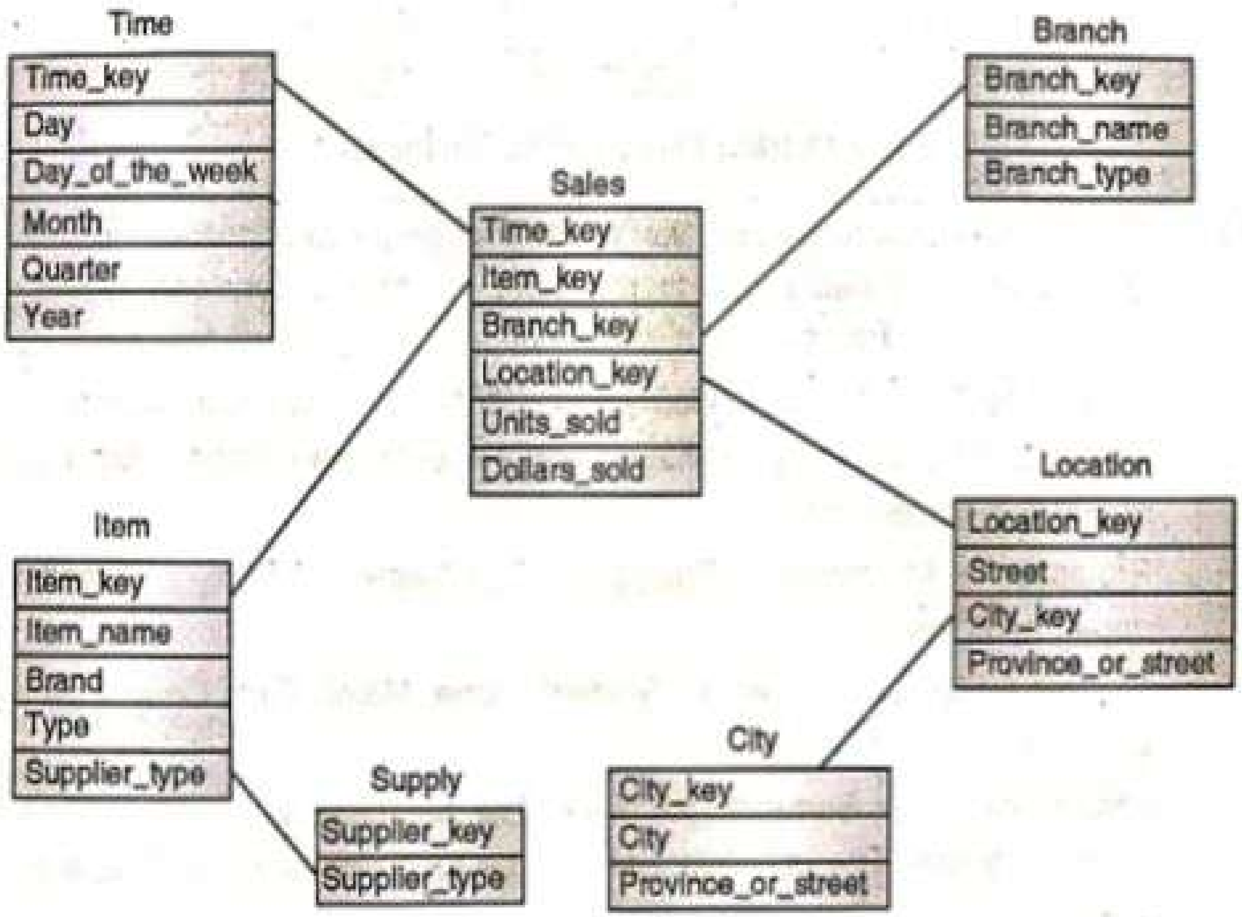
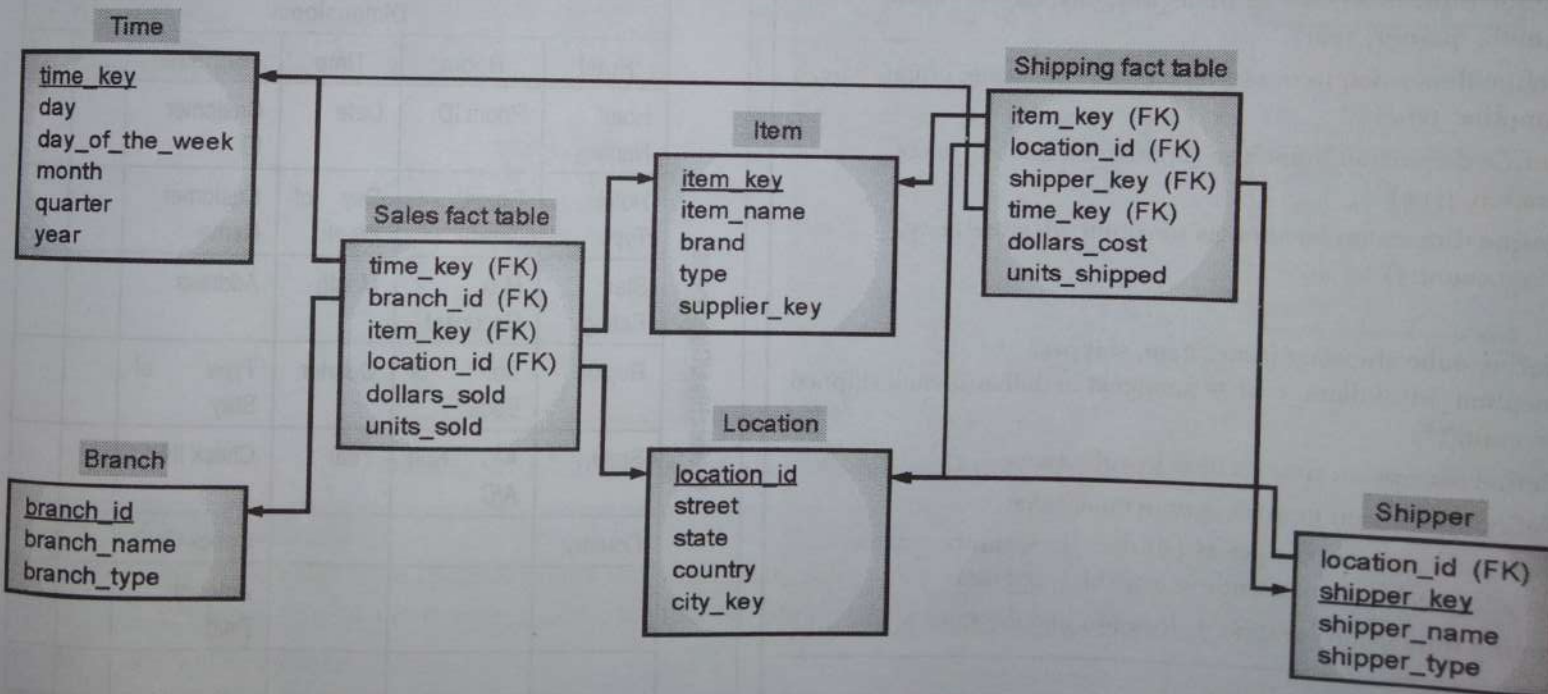


Fig. P. 1.17.1(a) : Sales Snowflake Schema

1.7.5(C) Example

A fact constellation schema is shown in Fig. 1.7.4 below. This schema specifies two fact tables, sales and shipping. The sales table definition is identical to that of the star schema above. The shipping table has four dimensions, or keys—item_key, time_key, shipper_key, location_id and two measures dollars_cost and units_shipped. A fact constellation schema allows dimension tables to be shared between fact tables. For example, the dimension tables for time, item, and location are shared between the sales and shipping fact tables.



(1A8)Fig. 1.7.4: Fact Constellation Schema for Digi1 Electronics Sale

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* OLAP operations :-

(1) Drill down
(2) Rollup
(3) Dice
(4) slice
(5) Pivot

(1) Drill drown / Rolldown

- Less detailed to Highhely detail.
- हर महिने का data अलग.

(2) Rollup / Drillup

- Highly detail to less detail.

(3) Dice

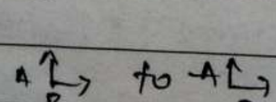
- Small piece of info. from chunk of info.

(4) slice

-
- slice is a subset of cuber.

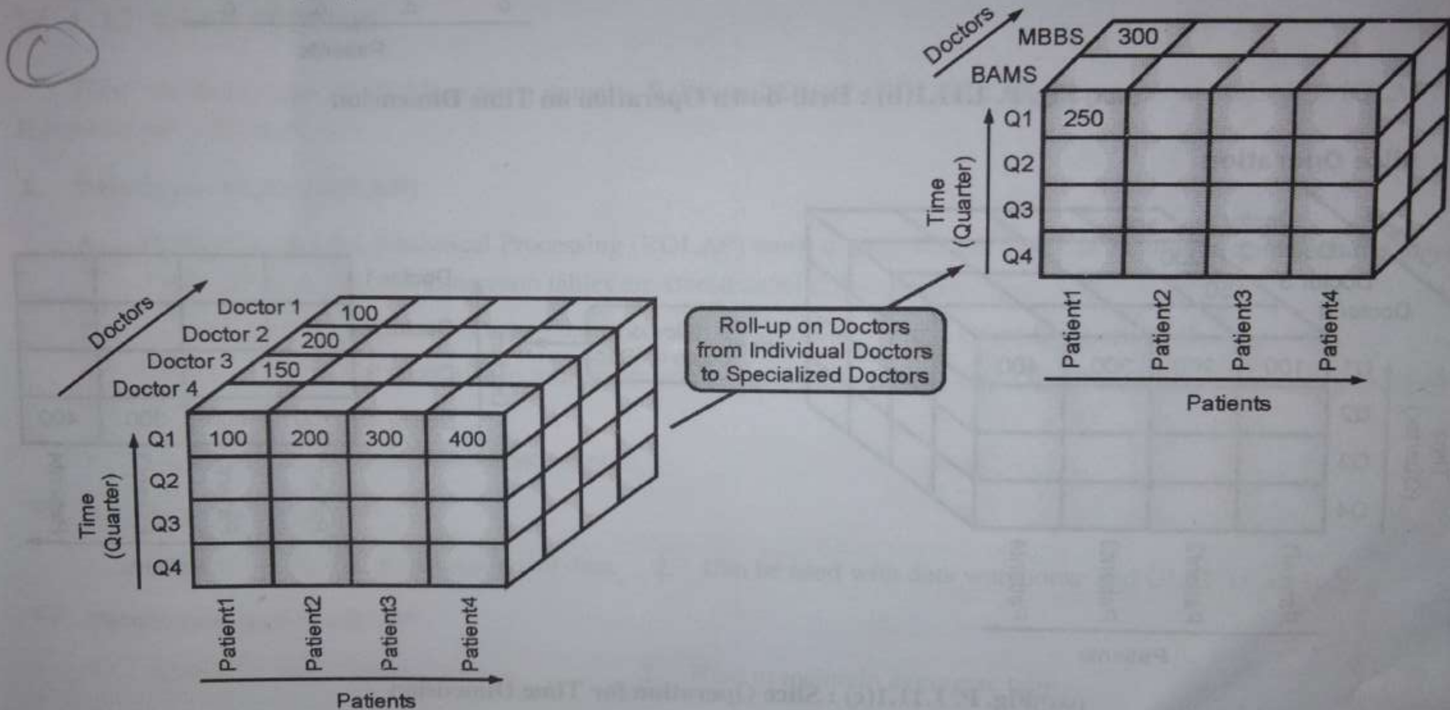
(5) Pivot / Rotation

- Rotate Current view to get New. View.



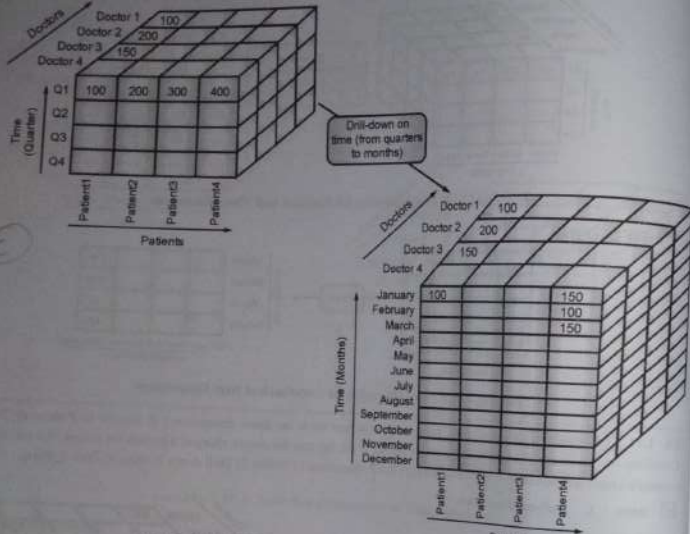
Ex. 1.11.1 : Consider a data warehouse for a hospital where there are three dimensions: a) Doctor b) Patient c) Time. Consider two measuresi) Countii) Charge where charge is the fee that the doctor charges a patient for a visit. For the above example create a cube and illustrate the following OLAP operations.1) Rollup 2) Drill down 3) Slice 4) Dice 5) Pivot.

☑ **Soln. : 1. Roll-up Operation**



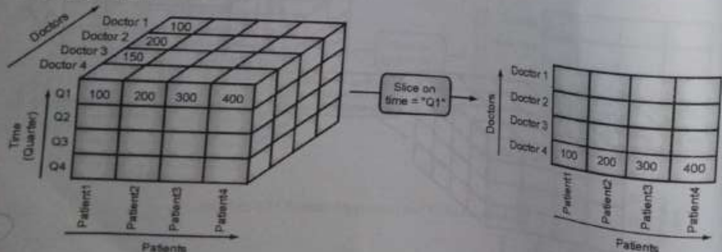
(1A25) **Fig. P. 1.11.1(a): Roll-up Operation on Doctors Dimension**

2. Drill-down Operation



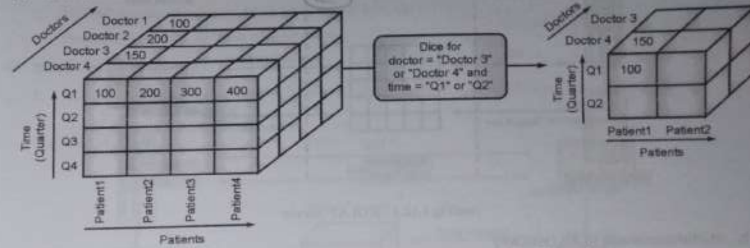
(1A26) Fig. P. 1.11.1(b) : Drill-down Operation on Time Dimension

3. Slice Operation

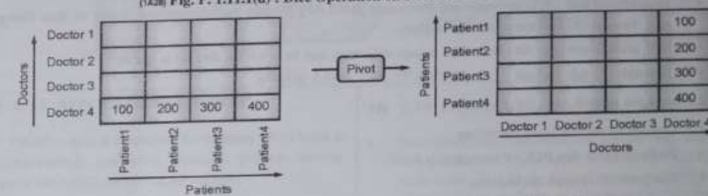


(1A27) Fig. P. 1.11.1(c) : Slice Operation for Time Dimension

4. Dice Operation



(1A28) Fig. P. 1.11.1(d) : Dice Operation on Doctors and Time Dimensions



(1A29) Fig. P. 1.11.1(e) : Pivot Operation on Doctors and Patients Dimensions

1.12 OLAP SERVERS

There are three types of OLAP servers, namely, Relational OLAP (ROLAP), Multidimensional OLAP (MOLAP) and Hybrid OLAP (HOLAP).

1. Relational OLAP (ROLAP)

- Relational On-Line Analytical Processing (ROLAP) work mainly for the data that resides in a relational database, where the base data and dimension tables are stored as relational tables.
- ROLAP servers are placed between the relational back-end server and client front-end tools.
- ROLAP servers use RDBMS to store and manage warehouse data, and OLAP middleware to support missing pieces.
- **Example :** DSS Server of Microstrategy

Advantages of ROLAP

1. ROLAP can handle large amounts of data.
2. Can be used with data warehouse and OLTP systems.

Disadvantages of ROLAP

1. Limited by SQL functionalities.
2. Hard to maintain aggregate tables.

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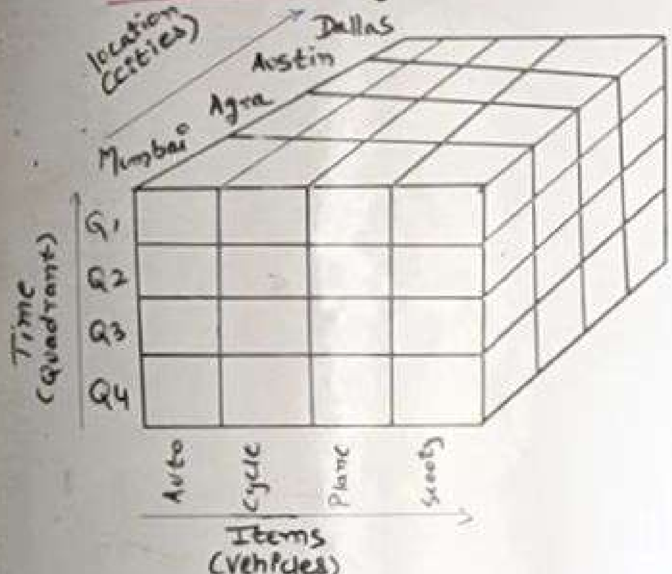
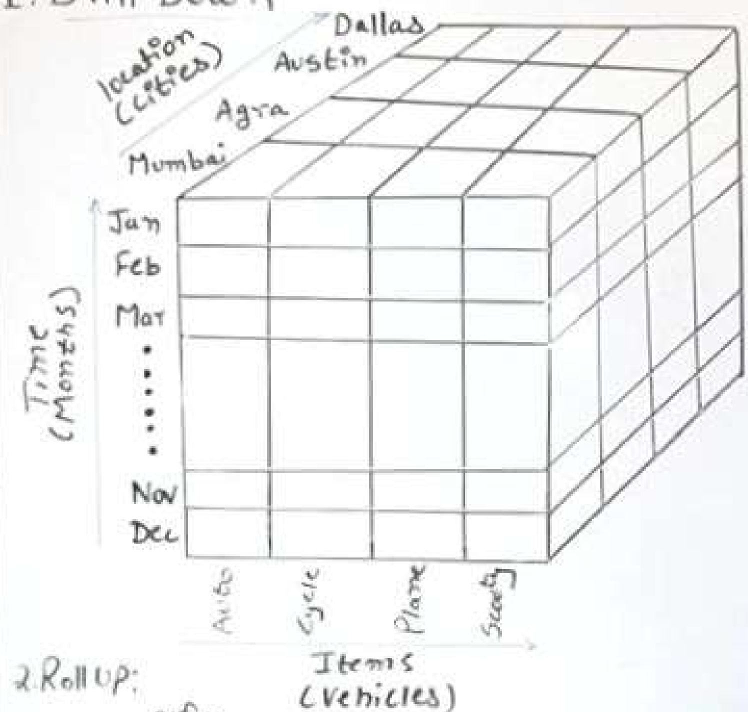
OLAP Operations

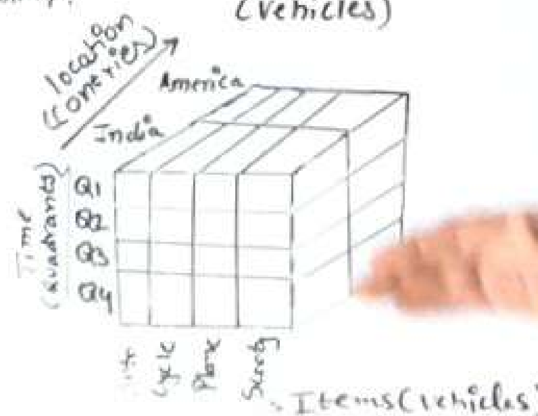
⇒ What is OLAP?



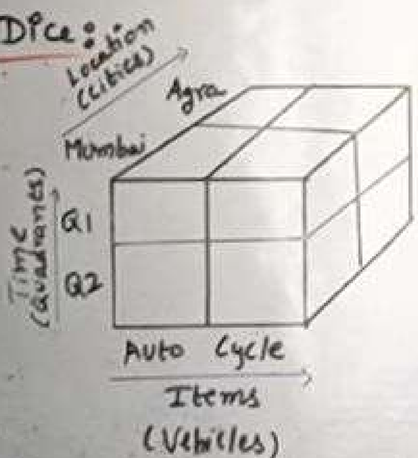
1. Drill Down



2. Roll UP:



3. Dice:



4.

5.

4. Slice

location (cities) \ Items (vehicles)	Auto	Cycle	Plane	Scooty
Dallas				
Austin				
Agra				
Mumbai				

5. Pivote

Items (vehicles) \ location (cities)	Dallas	Austin	Agra	Mumbai
Auto				
Cycle				
Plane				
Scooty				

Jan, Feb, Mar } Q1
Apr, May, June } Q2
July, Aug, Sep } Q3
Oct, Nov, Dec } Q4

OLAP Servers

* What are OLAP Servers?

× Types of OLAP servers:

1. Relational OLAP (ROLAP)

- It works with data that resides in Relational database.
- ROLAP Servers are placed between relational backend server & client frontend tools.

Advantages:

- Can handle huge amount of data
- Provides greater scalability
- Efficiently handles complex analytical queries which involves calculations and joins across multiple tables.

Disadvantages:

Only expertised person can handle ROLAP

2. Multidimensional OLAP (MOLAP):

- MOLAP helps dealing with Multidimensional data
- What is Multidimensional data?

Advantages:

- MOLAP is easy to use
- Information Retrieval is fast
- Suitable for slicing & Dicing operations

Disadvantages

- Can handle limited amount of data
- Can't deal with detailed data

3. Hybrid OLAP (HOLAP):

- HOLAP is combination of MOLAP & ROLAP
- Provides greater scalability (ROLAP) & Provides faster computation aswell (MOLAP).

Advantages:

HOLAP provides advantages of both: ROLAP & MOLAP

Disadvantages:

- Because of the support of ROLAP & MOLAP, HOLAP architecture is complex

* OLAP Servers :-

(1) Relational OLAP (ROLAP)
(2) Multidimensional OLAP (MOLAP)
(3) Hybrid OLAP (HOLAP)

* Updates to Dimension Table :-

- Important process in maintaining accurate & consistent data
- Occur when there are changes in real-world entities (like Address, price)
- Handled by SCD techniques (Slowly Changing Dimension).

Type 1 :- Overwrite Old Value.

- Old data in D.T. is written over with new value
- Simple & fast ; but loses prev. info.
  eg., Customer's City changes from "Mumbai" to "Pune".
- when Historical data is not imp; only current data is needed.

Type 2 :- Add New Record

- preserves historical data by adding new Row for every change.
- Additional field like Start-Date & End-Date or a Current-Flag
- F.T. can be linked to Current version using Surrogate keys.

eg.,

Cust-Id	Cust-Name	city	start-Date	end Date	current-flag
101	Riya	Mumbai	1Jan24	10Jun24	N
102	Riya	Pune	11Jun24	NULL	Y

- when historical tracking is important.

Type 3 :- Add New Attribute (column)

- keeps only limited history by adding new column to store prev. value.
- eg., column Prev-city = Mumbai ; Current-city = Pune.
- when tracking only recent change is reqd.

Type 4 :- History Table.

- Current data = Main table ; prev data = separate Historical table

Type 5 :- Hybrid Approach

- Type 1+2+3

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MODULE 6
Web mining.

1) What is Web Mining? Explain Technologies of Web Mining? UT2, MU SA, Top 12, Dec24, May23
2) What is Web Structure Mining? UT2, MU SA, T12, May25, Dec22, May23
3) Compare and contrast Datamining & Web Mining? UT2, Dec24
4) Explain Web content & Webpage mining in details? MU SA, T12, May25, Dec23, May24, May23
5) Is Web mining different from Classical data mining? MU SA, Dec22
6) Explain CLARANS extention in WebMining. MU SA, Dec22
7) Explain Page Rank Algorithm & techniques in the detail. Top 12, May25, Dec24, Dec23, May24, Dec22, May23, MU SA
8) Disscuss diff. appl of Web Mining. Dec24.
9) Types of Web Mining? Dec24, Dec22

Q) Compare and contrast data Mining & Web Mining.

Parameters	Data Mining	Web mining
Definition	Data Mining is the process that attempts to discover pattern & hidden knowledge in a large datasets in any system.	Web Mining is the process of data mining technique to automatically discover & extract info. from web documents.
Concept	Pattern identification from data available in any system.	Pattern identification from web data.
Categories	Clustering, Classification, regression, prediction, optimization and control.	Web content Mining, Web Structure Mining, Web usage Mining.
Access	Data Mining is access data privately.	Web Mining is access data publicly.
Dynamic	Data Mining is less dynamic.	Web mining is more dynamic.
Tools	Machine learning algo.	Scrapy, PageRank, Apache logs.
Who does?	Data Scientist & Engg.	Data Scientist / Analyst & engineers

* Compare : Data Mining & Web Mining :-

→ refer UT2 Ans.

* Web Mining :-

- Web Mining is an application of data Mining Techniques to find information patterns from the web data.
- Web Mining is the Process of Data Mining Techniques to automatically discover & extract information from Web Documents & services.
- The Goal of Web Mining is to look for patterns in Web data by collecting and analyzing information inorder to gain insight into Trends, the industry and users to general.
- Process of Web Mining:

Raw Data $\xrightarrow{\text{Mining tools}}$ Pattern $\xrightarrow[\text{\& Visualization}]{\text{Representation}}$ Knowledge

- Application of Webmining:

1) Web Mining helps to improve the power of Web Search engines such as Google, Yahoo, etc by classifying the Web documents and identifying the Web pages
2) Web Mining is used to predict user behavior.
3) Web Mining is very useful of a particular Website and e-service eg., landing page optimization.
4) Web Mining is very useful to e-commerce website and e-services.

6.11 Applications of Web Mining

1. E-Learning.
2. Digital libraries.
3. E –Government.
4. Electronic commerce.
5. E-Politics and E-Democracy.
6. Security and Crime Investigation.
7. Electronic Business.

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- ~~Types~~ / Techniques of Webmining:

① Web Continent Mining
② Web Structure Mining
③ Web Usage Mining

- Web Mining can be broadly divided into 3 different types

① Web Content Mining:
- Web Continent Mining is ~~responsible~~ responsible for finding proper & relatable information
- used for mining of useful data, info. and knowledge from webpages content.
- patterns through Web Crawling, Real data from the Web.
- eg:, Extracting project details from E-commerce websites.

② Web - Structure Mining:
- helps to find useful knowledge or information pattern from the structure of hyper links.
- patterns through nested webpages through hyperlink.
- eg:, Google's Page Rank Algorithm uses structure mining to rank ~~algorithm~~ webpages.

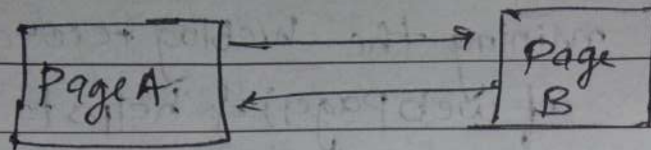
③ Web Usage Mining:
- is used for mining the Weblog records (access information of webpages) & helps to discover the user access patterns of web pages.
- patterns through WebServers logs.
- eg:, Analyzing user click patterns to improve websites design or recommended products.

* Web Mining different from Classical DM?

- Web Mining is similar to Data Mining.
- It differs in Data Collection.
- DataMining: The collection of data is already done and stored in a data Warehouse.
- Web Mining: Data Collection is done by crawling through a number of target Web pages.

* Page Rank Algorithm:-

- Page Rank Algorithm is applicable in Webpages.
- Page Rank is an Algorithm used by Google Search to rank websites in their search engine results.
- Page Rank was name after Larry Page, one of founders of Google.
- Each Web page is treated as a node in a directed graph, and hyperlinks represent edges between nodes.
- A page receives a higher rank if many other pages link to it and those linking pages themselves are Important.



Each pg has one outgoing link

$\therefore C_{out}(A) = 1$ & $C_{out}(B) = 1$

$\beta = 0.85$

$\therefore PR(A) = (1-\beta + \beta\ (PR(B)/1)$

$\therefore PR(B) = (1-\beta) + \beta\ (PR(B)/1)$

$\therefore PR(A) = 0.15 + 0.85 * 1 = 1$

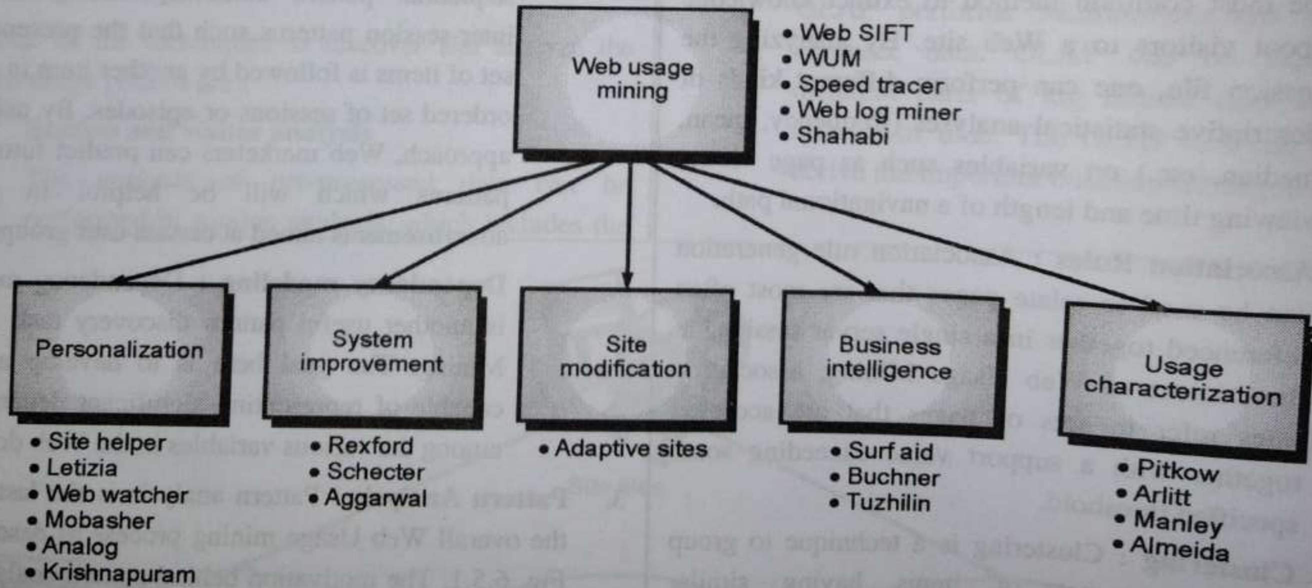
$\therefore PR(B) = 0.15 + 0.85 * 1 = 1.$

* CLARAN

Clustering large Appl^{n} based on Randomize search

6.5.1 Applications

1. Personalization for a user.
2. From frequent access behavior of user, overall performance can be improved.
3. Caching of frequently accessed pages.
4. Modifications of linkage structure, common access behavior are accessed.
5. Gather business intelligence to improve sales and advertisements.



(1F11)Fig. 6.5.2: Major application areas for Web Usage Mining

10 + 3 + 13 = 3|3   13 12 5 13 0

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MODULE 2

Intro to Data Mining, Data Exploration & Data Pre-processing.

1) Describe in details Data Mining Techniques UTI
2) Compare: Data Warehouse & Data Mining UTI
3) Explain different steps involved in Data preprocessing UTI, MU)A May23, Dec23, May25, T12
4) Describe steps involved in Data Mining when viewed as a process of knowledge discovery UTI, May23
5) Explain different Data Visualization Techniques UTI, MU)A, T12 Dec23, May25
6) Explain in detail the issue in Data Mining UTI, MU)A, T12, May25, May24.
7) Suppose that the data for analysis includes the attribute salary. We have the following values for the salary (in thousands of dollars), shown increasing order:
30, 36, 47, 50, 52, 52, 56, 60, 63, ~~70~~, 70, 110
i) What are the mean, Median, Mode & Mid Range of the data?
ii) First the first Quartile (Q1) & the third Quartile (Q3) of the data.
iii) Show the box plot of the data. UTI
8) Explain KDD process of Data Mining. MU)A, T12, Dec23, May24
9) Describe types of Data Mining MU)A
10) Discuss different types of Attributes MU)A. May25, Dec24, Dec22
11) Short note: Appln of Data mining. MU)A, Dec23, May24
12) Explain the foll. data Preprocessing Methods Dec24
i) Dimension Reduction ii) Data transformation & discretization
13) Explain the different techniques to handle noisy data.
14) Suppose a group of sales price records has been sorted as follows:
3, 7, 8, 13, 22, 22, 22, 26, 26, 28, 30, 37.
Partition them into 3 bins by equal freq. (Equi-depth) partitioning method. Perform data smoothing by bin mean & bin boundary. Dec24.

14) Explain in brief What is data Discretization & Concept Hierarchy. May 24

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* Answers :-

Compare : Data Warehouse & Data Mining

Data Warehouse	Data Mining
i) Data Warehousing is a method of Centralizing data from different sources into one common repository.	i) Data mining is a method of Comparing large amounts of data to finding right patterns.
ii) Must take place before Data Mining.	ii) After Data Warehousing process
iii) Carried out by Engineers	iii) Business users with help of Engineers.
iv) Data is saved on a regular basis.	iv) Data is analyzed on a regular basis
v) D.W. shows what has happened.	v) D.M. explains why & what will happen.
vi) DW use for OLAP operation	vi) DM used for Algo. prediction & classificatn.
vii) DW focuses on Storage & management.	vii) D.M. focuses on Analysis and discovery.
viii) To arrange data for easy retrieval, reporting & analysis	viii) To discover knowledge & support decision making.
ix) Eg., stores last 5yr sales.	ix) Eg., ~~stores to~~ predicts last 5yr sales.

Q7 Suppose that the data for analysis includes the attribute salary. We have the foll. values for salary (in thousands of dollars) shown in increasing order:
30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110.
i) what are Mean, Median, Mode & Midrange of data?
ii) find 1st quartile (Q1) & third quartile (Q3) of data.
iii) Show the boxplot of the data.

→ We have salary data (in thousands of dollars):
30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110.

i) Mean, Median, Mode & Midrange?

a) Mean $= \frac{x_1 + x_2 + \dots + x_n}{n}$ $\frac{\text{(sum of all values)}}{\text{(count)}}$

$= \frac{696}{12}$

∴ $\boxed{\text{Mean} = 58}$

b) Median = mean of middle 2 values of set

$= \frac{(52+56)}{2}$

∴ $\boxed{\text{Median} = 54}$

c) Mode (Most frequent values)
Here 52, 70. both appears twice.

∴ $\boxed{\text{Mode} = 52, 70}$

d) Midrange $= \frac{(\min + \max)}{2}$

$= \frac{30+110}{2}$

∴ $\boxed{\text{Median} = 70}$

ii) Quartiles

∴ Q_1 = Median of lower half {30, 36, 47, 50, 52, 52}

$= \frac{47+50}{2}$

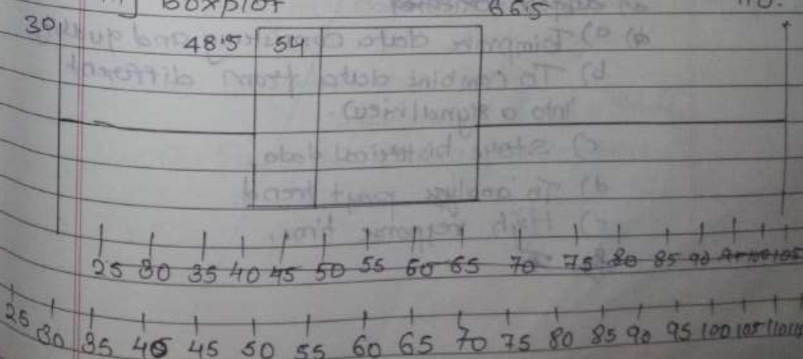
∴ $\boxed{Q_1 = 48.5}$

∴ Q_3 = Median of Upper half {56, 60, 63, 70, 70, 110}

$= \frac{63+70}{2}$

∴ $\boxed{Q_3 = 66.5}$

iii) Boxplot



Suppose that the data for analysis includes the attribute age. The age values for data tuples are (in increasing order)
{13,15,1[illegible],16,19,20,20,21,22,22,25,25,25,25,
30,3[illegible],35,35,35,35,36,40,45,46,52,70}

[illegible]an, Median, Mode, Midrange, Range, [illegible]les (Q1,Q2,Q3), Give five number [illegible]mmary of the data & Show a box Plot of [illegible] data.

1. Mean $= \sum_{i=1}^{n} \frac{x_i}{n} = \frac{809}{27} = 30$

2. Median $= \left(\frac{N+1}{2}\right)^{th}$ term

$= \left(\frac{27+1}{2}\right)^{th}$ term

$= \left(\frac{28}{2}\right)^{th}$ term

$= 14^{th}$ term

$= 25$

3. Mode: Since two numbers are repeating highest no. of times So the dataset is a bimodel dataset. Modes of the dataset are 25 & 35.

4. Midrange: $\frac{13+70}{2}$

$= 41.5$

5. Range: $70 - 13$

$= 57$

6. Quartile 2 (Q2) = Median

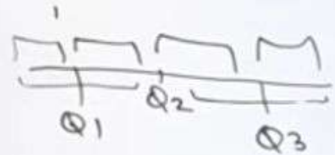
$= 25$

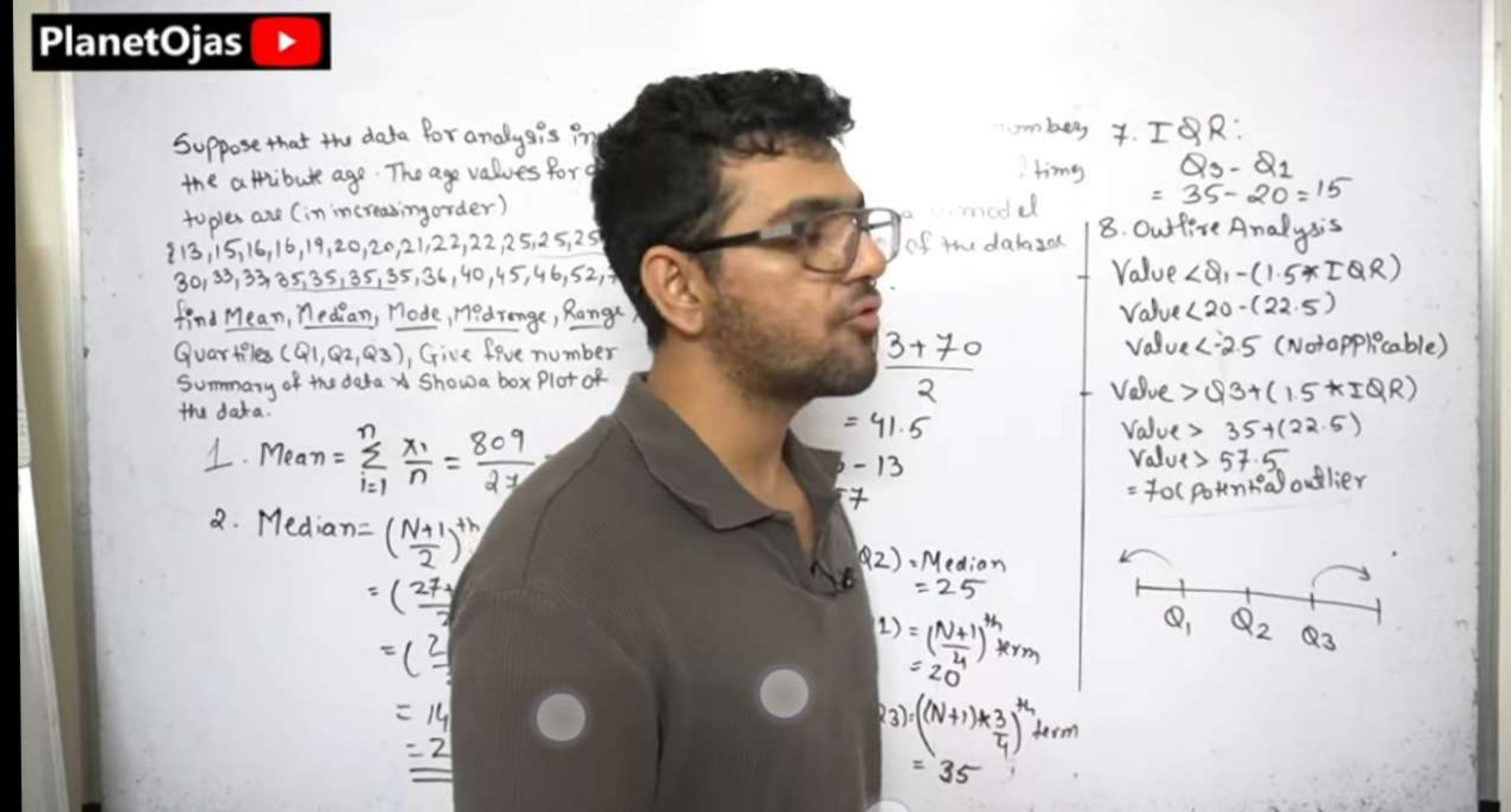
Quartile 1 (Q1) $= \left(\frac{N+1}{4}\right)^{th}$ term

$= 20$

Quartile 3 (Q3) $= \left((N+1) * \frac{3}{4}\right)^{th}$ term

$=$




PlanetOjas
Suppose that the data for analysis in ... number
the attribute age. The age values for ... times
tuples are (in increasing order)
{13,15,16,16,19,20,20,21,22,22,25,25,25
30,33,33,35,35,35,35,36,40,45,46,52,
find Mean, Median, Mode, Midrange, Range
Quartiles (Q1,Q2,Q3), Give five number
Summary of the data & Show a box Plot of
the data.
1. Mean = $\sum_{i=1}^{n} \frac{x_i}{n} = \frac{809}{27}$
2. Median = $\left(\frac{N+1}{2}\right)^{th}$
= 41.5
- 13
(Q2) = Median
= 25
(Q1) = $\left(\frac{N+1}{4}\right)^{th}$ term
= 20
(Q3) = $\left((N+1) * \frac{3}{4}\right)^{th}$ term
= 35
7. IQR:
Q3 - Q1
= 35 - 20 = 15
8. Outlire Analysis
Value < Q1 - (1.5 * IQR)
Value < 20 - (22.5)
Value < -2.5 (Not applicable)
Value > Q3 + (1.5 * IQR)
Value > 35 + (22.5)
Value > 57.5
= 70 (potential outlier)
Q1
Q2
Q3

Q] Describe in details Data Mining Techniques.

→ Data mining is the process of discovering the patterns trends and useful information from large data sets.

Different techniques are used depending on the type of data and the goal.

Data mining Techniques

Classification | Clustering | Regression | Association | Detection | Prediction

1) Classification :-

Data can be divided into Predefined grps or classes.

For eg; emails can be classified as spams or not spams.

2) Clustering :-

Clustering is very similarly to classificatn. Similar data objects are grouped together without predefined labels.

eg; ~~customer base~~ customers can be grouped based on their buying behaviour.

3) Regression :-

One data keep reln betn 2nd data but it's not necessary to One data is depending on 2nd and 2nd also depending on 1st. These techniques predict Continuous Values.

4) Association Rule mining :-
It finds relationships betn items in a dataset ; more specifically to dependently linked variables.
for eg., if customer buys bread, they are also likely to buy butter.

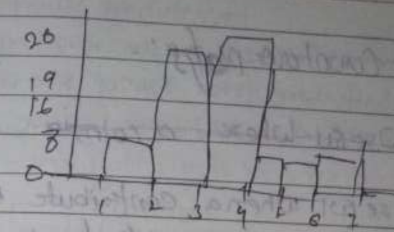
5) Detection :-
Used to find unusual data that does not follow the normal pattern.
for eg., detecting ; Froaud in credit card transaction.

6) Prediction :-
Past data is used to Forecast future outcomes. For eg., predicted next month sales.

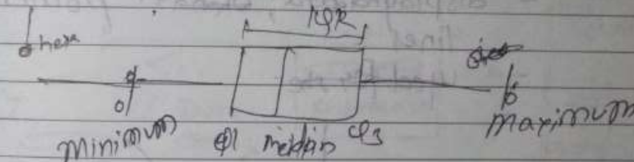
Q; Explain different data visualizn techniques.
→ Data visualization is the process of representing data in a graphical or visual format so that patterns, trends and insights can be easily understood.

Visualization Techniques :-

1) Histogram :
- Usually shows the distribution of values of a single variable.
- eg., petalwith.



2) Boxplots :-
- Invented by J. Turkey
= Another way of displaying ~~the~~ distribution of a data.
- Use to compare attributes.



3) Scarlot plots :-
- Attributes values determine the position.
- 2D and 3 both available

: Scatter plot

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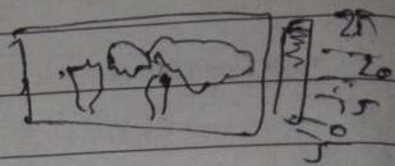
4) ~~Countour~~ plots:-

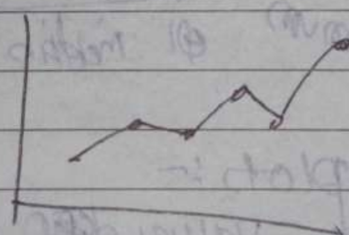


- ~~Useful where a column~~

- useful when a ~~contribute~~ ~~[illegible]~~ attribute is measured on a spatial grip.
- eg:, Candle play, temp., rainfall, air pressure.

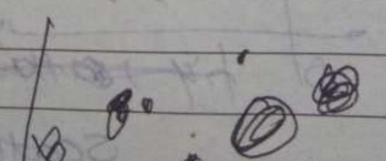
5) Line chart:-

- displayed data; ~~when a~~ points connected by lines
- ~~used for the~~.



6) Bubble chart

- ~~Usefu for Multivariball~~,
- useful for multi variable comparison



* Compare :- Data Warehouse & Data Mining :-

→ refer to UT1 Ans

* Application of Data Mining :-

- Data Mining is widely ~~used to~~ applied in multiple domains to extract hiddens patterns and useful information from large datasets.

* Data Minining Architecture :-

① Datasoure
② Data Warehouse Server
③ Data Machine Engine
④ Pattern Evaluation Module
⑤ GUI

* Appln of Data Mining :

① Mobile service Providers
② Artificial Intelligence
③ Ecommerce.
④ Crime preventn
⑤ Research
⑥ Farming
⑦ Insurance

...cal ...ng ...he ...rs ...to ...ta ...ct	Data warehouse allows users to access critical data from the number of sources in a single place. Therefore, it saves user's time of retrieving data from multiple sources.
...o ...e ...a	Once you input any information into Data warehouse system, you will unlikely to lose track of this data again. You need to conduct a quick search, helps you to find the right statistic information.

TASK PRIMITIVES

...mining task in mind, that is, ...that he or she would like to

...specified in the form of a ...is input to the data mining

...ed in terms of data mining

...he user to interactively ...a mining system during ...t the mining process, or ...fferent angles or depths.

...g Task Primitives.

...ata to be mined : This ...atabase or the set of data ...sted. This includes the ...arehouse dimensions of ...relevant attributes or

...mined : This specifies ...be performed, such as ...on, association or ...ification, prediction, ...olution analysis.

...domain ...to be mined is useful for guiding the knowledge discovery process and for evaluating the patterns found. Concept hierarchies are a popular form of background knowledge, which allow data to be mined at multiple levels of abstraction.

4. **The interestingness measures and thresholds for pattern evaluation :** They may be used to guide the mining process or, after discovery, to evaluate the discovered patterns. Different kinds of knowledge may have different interestingness measures. For example, interestingness measures for association rules include support and confidence. Rules whose support and confidence values are below user-specified thresholds are considered uninteresting.

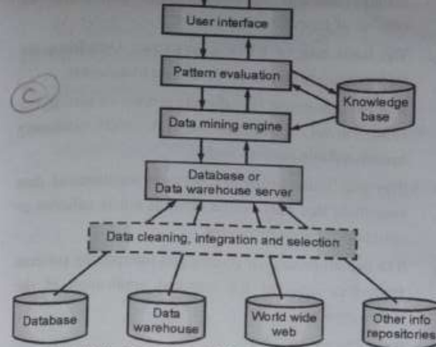
5. **The expected representation for visualizing the discovered patterns :** This refers to the form in which discovered patterns are to be displayed, which may include rules, tables, charts, graphs, decision trees, and cubes.

2.3 DATA MINING ARCHITECTURE

The significant components of data mining systems are a data source, data mining engine, data warehouse server, the pattern evaluation module, graphical user interface, and knowledge base.

Data Source

- The actual source of data is the Database, data warehouse, World Wide Web (WWW), text files, and other documents. You need a huge amount of historical data for data mining to be successful.
- Organizations typically store data in databases or data warehouses. Data warehouses may comprise one or more databases, text files spreadsheets, or other repositories of data.
- Sometimes, even plain text files or spreadsheets may contain information. Another primary source of data is the World Wide Web or the internet.



(1B1)Fig. 2.3.1: Data Mining Architecture

Data Cleaning, Integration and Selection

- Before passing the data to the database or data warehouse server, the data must be cleaned, integrated and selected.
- As the information comes from various sources and in different formats, it can't be used directly for the data mining procedure because the data may not be complete and accurate. So, the first data requires to be cleaned and unified.
- More information than needed will be collected from various data sources, and only the data of interest will have to be selected and passed to the server. These procedures are not as easy as we think. Several methods may be performed on the data as part of selection, integration, and cleaning.

Database or Data Warehouse Server

The database or data warehouse server consists of the original data that is ready to be processed. Hence, the server is cause for retrieving the relevant data that is based on data mining as per user request.

Data Mining Engine

- The data mining engine is a major component of any data mining system. It contains several modules for operating data mining tasks, including association, characterization, classification, clustering, prediction, time-series analysis, etc.
- In other words, we can say data mining is the root of our data mining architecture. It comprises instruments and software used to obtain insights and knowledge from data collected from various data sources and stored within the data warehouse.

Pattern Evaluation Module

- The Pattern evaluation module is primarily responsible for the measure of investigation of the pattern by using a threshold value. It collaborates with the data mining engine to focus the search on exciting patterns.
- This segment commonly employs stake measures that cooperate with the data mining modules to focus the search towards fascinating patterns. It might utilize a stake threshold to filter out discovered patterns.
- On the other hand, the pattern evaluation module might be coordinated with the mining module, depending on the implementation of the data mining techniques used. For efficient data mining, it is abnormally suggested to push the evaluation of pattern stake as much as possible into the mining procedure to confine the search to only fascinating patterns.

Graphical User Interface

- The graphical user interface (GUI) module communicates between the data mining system and the user. This module helps the user to easily and efficiently use the system without knowing the complexity of the process.
- This module cooperates with the data mining system when the user specifies a query or a task and displays the results.

Knowledge Base

- The knowledge base is helpful in the entire process of data mining. It might be helpful to guide the search or evaluate the stake of the result patterns.
- The knowledge base may even contain user views and data from user experiences that might be helpful in the data mining process.
- The data mining engine may receive inputs from the knowledge base to make the result more accurate and reliable.
- The pattern assessment module regularly interacts with the knowledge base to get inputs, and also update it.

that, the user can withdraw that information effectively from the large amount of data in various databases.

- **Parallel, distributed and incremental mining algorithm :** There are a lot factors which can be responsible for the development of parallel and distributed algorithms in data mining. These factors are large in size of database, huge distribution of data, and data mining method that are complex. In this process, in the first and foremost step, the algorithm divides the data from database into various partition. In the next step, that data is processed such that it is situated in parallel manner. Then in the last step, the result from the partition is merged.

3. Diverse Data Types Issues

The issues in this category are given below:

- **Handling of relational and complex types of data :** The database may contain the various data objects. For example, complex, multimedia, temporal data, or spatial data objects. It is very difficult to mine all these data with the help of a single system.
- **Mining information from heterogeneous databases and global information systems :** The problem in this kind of issue is to mine the knowledge from various data sources. These data are not available as a single source instead these data are available at the different data sources on LAN or WAN. The structures of these data are different as well.

2.6 APPLICATIONS OF DATA MINING

The importance of data mining and analysis is growing day by day in our real life. Today most organizations use data mining for analysis of Big Data. Let us see how this technology benefit different users.

1. Mobile Service Providers

- Mobile service providers use data mining to design their marketing campaigns and to retain customers from moving to other vendors.
- From a large amount of data such as billing information, email, text messages, web data transmissions, and customer service, the data mining tools can predict "churn" that tells the customers who are looking to change the vendors.
- With these results, a probability score is given. The mobile service providers are then able to provide incentives, offers to customers who are at higher risk of churning. This kind of mining is often used by major service providers such as broadband, phone, gas providers, etc.

2. Retail Sector

- Data Mining helps the supermarket and retail sector owners to know the choices of the customers. Looking at the purchase history of the customers, the data mining tools show the buying preferences of the customers.
- With the help of these results, the supermarkets design the placements of products on shelves and bring out offers on items such as coupons on matching products, and special discounts on some products.
- These campaigns are based on RFM grouping. RFM stands for recency, frequency, and monetary grouping. The promotions and marketing campaigns are customized for these segments. The customer who spends a lot but very less frequently will be treated differently from the customer who buys every 2-3 days but of less amount.
- Data Mining can be used for product recommendation and cross-referencing of items.

3. Artificial Intelligence

- A system is made artificially intelligent by feeding it with relevant patterns. These patterns come from data mining outputs. The outputs of the artificially intelligent systems are also analysed for their relevance using the data mining techniques.
- The recommender systems use data mining techniques to make personalized recommendations when the customer is interacting with the machines. The artificial intelligence is used on mined data such as giving product recommendations based on the past purchasing history of the customer in Amazon.

4. E-commerce

- Many E-commerce sites use data mining to offer cross-selling and upselling of their products. The shopping sites such as Amazon, Flipkart show "People also viewed", "Frequently bought together" to the customers who are interacting with the site.

- These recommendations are provided using data mining over the purchasing history of the customers of the website.

5. Science and Engineering

- With the advent of data mining, scientific applications are now moving from statistical techniques to using "collect and store data" techniques, and then perform mining on new data, output new results and experiment with the process. A large amount of data is collected from scientific domains such as astronomy, geology, satellite sensors, global positioning system, etc.
- Data mining in computer science helps to monitor system status, improve its performance, find out software bugs, discover plagiarism and find out faults. Data mining also helps in analyzing the user feedback regarding products, articles to deduce opinions and sentiments of the views.

6. Crime Prevention

- Data Mining detects outliers across a vast amount of data. The criminal data includes all details of the crime that has happened. Data Mining will study the patterns and trends and predict future events with better accuracy.
- The agencies can find out which area is more prone to crime, how much police personnel should be deployed, which age group should be targeted, vehicle numbers to be scrutinized, etc.

7. Research

Researchers use Data Mining tools to explore the associations between the parameters under research such as environmental conditions like air pollution and the spread of diseases like asthma among people in targeted regions.

8. Farming

Farmers use Data Mining to find out the yield of vegetables with the amount of water required by the plants.

9. Automation

By using data mining, the computer systems learn to recognize patterns among the parameters which are under comparison. The system will store the patterns that will be useful in the future to achieve business goals. This learning is automation as it helps in meeting the targets through machine learning.

10. Dynamic Pricing

Data mining helps the service providers such as cab services to dynamically charge the customers based on the demand and supply. It is one of the key factors for the success of companies.

11. Transportation

Data Mining helps in scheduling the moving of vehicles from warehouses to outlets and analyze the product loading patterns.

12. Insurance

Data mining methods help in forecasting the customers who buy the policies, analyze the medical claims that are used together, find out fraudulent behavior and risky customers.

2.7 DATA EXPLORATION

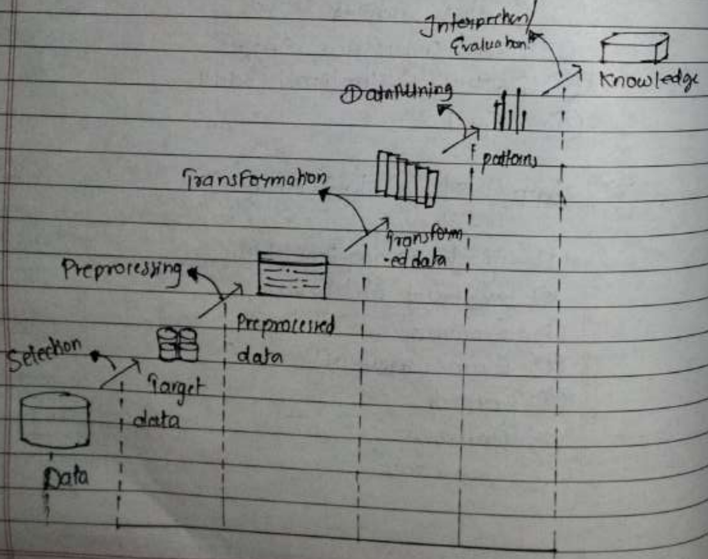
- Data exploration refers to the initial step in data analysis in which data analysts use data visualization and statistical techniques to describe dataset characterizations, such as size, quantity, and accuracy, in order to better understand the nature of the data.
- Data exploration techniques include both manual analysis and automated data exploration software solutions that visually explore and identify relationships between different data variables, the structure of the dataset, the presence of outliers, and the distribution of data values in order to reveal patterns and points of interest, enabling data analysts to gain greater insight into the raw data.
- Data is often gathered in large, unstructured volumes from various sources and data analysts must first understand and develop a comprehensive view of the data before extracting relevant data for further analysis, such as univariate, bivariate, multivariate, and principal components analysis.

* KDD Process :- / steps In Data Mining :-

pre-processing
Data mining
Post processing

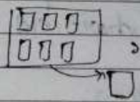
- "Knowledge Discovery in Database" (KDD) is the process of searching for hidden knowledge in the massive amount of data that we are technically capable of generating and storing.

- The Basic Task of KDD is to extract knowledge (or information) from Lower level data (d B).



- The overall process of finding & interpreting patterns from data involves the repeated appln of the following steps :

(1) Data selection :-
Collecting only necessary information to the model.



(2) Data cleaning :
Removal of noise, incosistent data
↳ यहा फ़िर datamining apply नहीं हो सकती → for avoid inaccuracy

(3) Data Integration:
Data which is available from Multiple Data Sources may be combined into single standard format.

(4) Data Transformation:
Data are transformed into forms appropriate for Mining

(5) Data mining :
The preprocessed data which is now ready for Data mining on which intelligent methods are applied to extract data pattern

(6) Pattern Evaluation:
To identify the reqrd hidden patterns, such there evaluation is done

(7) Knowledge Presentation / visualizn :
Visualization and knowledge presentation techniques are used to present it in particular form (graph, charts, etc) as per userneed.

* Issues in data Mining:

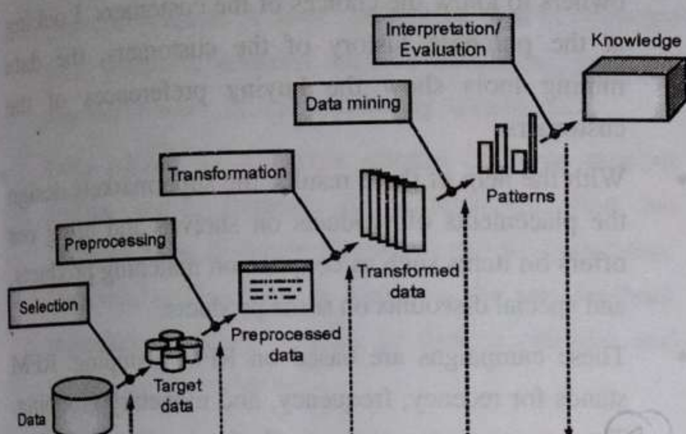
(1) Mining Methodology & User Interaction
(2) Performance Issue
(3) Issue related to diversity of database types

6. Pattern Evaluation

To identify the truly interesting patterns representing knowledge based on interesting measures.

7. Knowledge Presentation

- Visualization and knowledge representation techniques are used to present mined knowledge to users.
- Visualizations can be in form of graphs, charts or table.

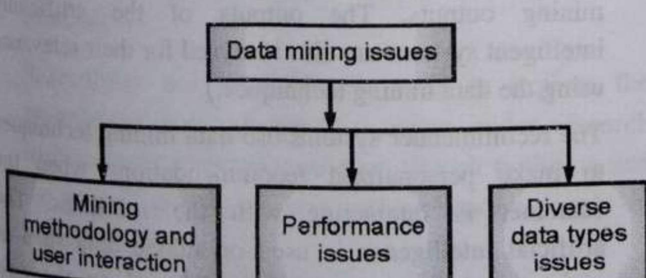


(1B2)Fig. 2.4.1: KDD Process

2.5 ISSUES IN DATA MINING

Data mining systems face a lot of challenges and issues in today's world. Some of them are:

1. Mining methodology and user interaction issues
2. Performance issues
3. Issues relating to the diversity of database types



(1B3)Fig. 2.5.1 : Data Mining Issues

1. Mining Methodology and User Interaction Issues

The issues in this category are as follows:

- **Mining different kinds of knowledge in databases :** This issue is responsible for addressing the problems of covering a big range of data in order to meet the needs of the client or the customer. Due to the different information or a different way, it becomes difficult for a user to cover a big range of knowledge discovery task.
- **Interactive mining of knowledge at multiple levels of abstraction :** Interactive mining is very crucial because it permits the user to focus the search for patterns, providing and refining data mining requests based on the results that were returned. In simpler words, it allows user to focus the search on patterns from various different angles.
- **Incorporation of background of knowledge :** The main work of background knowledge is to continue the process of discovery and indicate the patterns or trends that were seen in the process. Background knowledge can also be used to express the patterns or trends observed in brief and precise terms. It can also be represented at different levels of abstraction.
- **Data mining query languages and ad hoc data mining :** Data Mining Query language is responsible for giving access to the user such that it describes ad hoc mining tasks as well and it needs to be integrated with a data warehouse query language.
- **Presentation and visualization of data mining results:** In this issue, the patterns or trends that are discovered are to be rendered in high level languages and visual representations. The representation has to be written so that it is simply understood by everyone.
- **Handling noisy or incomplete data:** For this process, the data cleaning methods are used. It is a convenient way of handling the noise and the incomplete objects in data mining. Without data cleaning methods, there will be no accuracy in the discovered patterns. And then these patterns will be poor in quality.

2. Performance Issues

It has been noticed several times there are performance related issues in data mining as well. These issues are as follows:

- **Efficiency and Scalability of data mining algorithm :** Efficiency and Scalability is very important when it comes to data mining process. It is also very necessary because with the help of using this, the user can withdraw the information from the data in a more effective and productive manner. On top of that, the user can withdraw that information effectively from the large amount of data in various databases.

- **Parallel, distributed and incremental mining algorithm :** There are a lot factors which can be responsible for the development of parallel and distributed algorithms in data mining. These factors are large in size of database, huge distribution of data, and data mining method that are complex. In this process, in the first and foremost step, the algorithm divides the data from database into various partition. In the next step, that data is processed such that it is situated in parallel manner. Then in the last step, the result from the partition is merged.

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The issues in this category are given below:

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Ex. 2.12.1 : Suppose a group of age records has been sorted as follows: 3, 7, 8,13, 22, 22, 22, 26, 26, 28, 30, 37. Partition them into three bins by equal-frequency (Equi-depth) partitioning method. Perform data smoothing by bin mean, and bin boundary.

☑ **Soln. :**

Smooth the data by equal frequency bins

Given three bins. There are total 12 observations. Hence, by equi-depth partitioning method, each bin will have 4 observations.

Modu

2

- Bin 1: 3, 7, 8, 13
- Bin 2: 22, 22, 22, 26
- Bin 3: 26, 28, 30, 37

5 avg of this

37-3 = 34 ÷ 3 / 14...

3 | 34 / 3 / 14

Smooth the data by bin mean

We take average of each bin and replace each data value by mean value in corresponding bin.

3+14=17

18+14=32

33+14=47

- Bin 1: 8, 8, 8, 8
- Bin 2: 23, 23, 23, 23
- Bin 3: 30, 30, 30, 30

Smooth the data by bin boundary

We take difference of each data value and the bin boundaries. Each bin value is then replaced by the closest boundary value.

- Bin 1: 3, 3, 3, 13
- Bin 2: 22, 22, 22, 26
- Bin 3: 26, 26, 26, 37

Ex. 2.12.2 MU - June 2021

ppose a group of sales price records has been sorted as llows: 6, 9, 12, 13, 15, 25, 50, 70, 72, 92, 204, 232. artition them into three bins by equal-frequency (Equi-epth) partitioning method. Perform data smoothing by bin ean.

☑ **Soln. :**

Smooth the data by equal frequency bins

Given three bins. There are total 12 observations. Hence, by equi-depth partitioning method, each bin will have 4 observations.

- Bin 1: 6, 9, 12, 13
- Bin 2: 15, 25, 50, 70
- Bin 3: 72, 92, 204, 232

Smooth the data by bin mean

We take average of each bin and replace each data value by mean value in corresponding bin.

- Bin 1: 10, 10, 10, 10
- Bin 2: 40, 40, 40, 40
- Bin 3: 150, 150, 150, 150

Data Smoothing & Data Reduction

Suppose a group of age records has been sorted as follows:
{5, 12, 13, 14, 15, 18, 26, 28, 42}
Perform data smoothing & data reduction.

=> Equal freq Bin:

Bin1: {5, 12, 13}
Bin2: {14, 15, 18}
Bin3: {26, 28, 42}

Equal Width Bin:

Bin1: {5, 12, 13, 14, 15}
Bin2: {18, 26, 28}
Bin3: {42}

* Smoothing by Mean (1):

Bin1: {10, 10, 10}
Bin2: {16, 16, 16}
Bin3: {32, 32, 32}

* Smoothing by Boundry (1):

Bin1: {5, 13, 13}
Bin2: {14, 14, 18}
Bin3: {26, 26, 42}

* Smoothing by Mean (2):

1. Equal frequency Bin
2. Equal Width Bin
3. Smoothing by Mean (1)
4. Smoothing by boundry (1)
5. Smoothing by Mean (2)
6. Smoothing by boundry (2)

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Data Smoothing & Data Reduction

...of age records
...s follows:
...6, 28, 42}
...ng &

...Bin:

{5, 12, 13}
...4, 15, 18}
...28, 42}

...n:

...15}

* Smoothing by Mean (1):
  - Bin 1: {10, 10, 10}
  - Bin 2: {16, 16, 16}
  - Bin 3: {32, 32, 32}
* Smoothing by Boundry (1):
  - Bin 1: {5, 13, 13}
  - Bin 2: {14, 14, 18}
  - Bin 3: {26, 26, 42}
* Smoothing by Mean (2):
  - Bin 1: {12, 12, 12, 12, 12}
  - Bin 2: {24, 24, 24}
  - Bin 3: {42}

1. Equal frequency Bin
2. Equal Width Bin
3. Smoothing by Mean (1)
4. Smoothing by boundry (1)
5. Smoothing by Mean (2)
6. Smoothing by boundry (2)

Smoothing by boundry (2)
- Bin 1: {5, 15, 15, 15, 15}
- Bin 2: {18, 28, 28}
- Bin 3: {42}

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4.1.2 Cluster Analysis

- Cluster Analysis in data mining means that to find out the group of objects which are similar to each other in the group but are different from the object in other groups.
- A good clustering algorithm aims to obtain clusters whose:

  (a) The intra-cluster similarities are high. It implies that the data present inside the cluster is similar to one another.

  (b) The inter-cluster similarity is low. It means each cluster holds data that is not similar to other data.

4.1.3 Applications of Cluster Analysis

- In many applications, clustering analysis is widely used, such as data analysis, market research, pattern recognition, and image processing.
- It assists marketers to find different groups in their client base and based on the purchasing patterns. They can characterize their customer groups.
- It helps in allocating documents on the internet for data discovery.
- Clustering is also used in tracking applications such as detection of credit card fraud.
- As a data mining function, cluster analysis serves as a tool to gain insight into the distribution of data to analyze the characteristics of each cluster.
- In terms of biology, it can be used to determine plant and animal taxonomies, categorization of genes with the same functionalities and gain insight into structure inherent to populations.
- It helps in the identification of areas of similar land that are used in an earth observation database and the identification of house groups in a city according to house type, value, and geographical location.

Ex. 4.4.1 : Using k-means clustering, cluster the following data into two clusters and show each step. QB2
{2, 4, 10, 12, 3, 20, 30, 11, 25}

☑ **Soln. :**

- **Step 1 : Randomly partition the given data set into two clusters and find the cluster mean.**

  $k_1 = \{2, 3\} \rightarrow m_1 = 2.5$

  $k_2 = \{4, 10, 12, 20, 30, 11, 25\} \rightarrow m_2 = 16$

- **Step 2 : Reassign the data points to each cluster and find again the cluster mean.**

  $k_1 = \{2, 3, 4\} \rightarrow m_1 = 3$

  $k_2 = \{10, 12, 20, 30, 11, 25\} \rightarrow m_2 = 18$

- **Step 3 : Reassign and find cluster mean.**

  $k_1 = \{2, 3, 4, 10\} \rightarrow m_1 = 4.75$

  $k_2 = \{12, 20, 30, 11, 25\} \rightarrow m_2 = 19.6$

- **Step 4 : Reassign and find cluster mean.**

  $k_1 = \{2, 3, 4, 10, 11, 12\} \rightarrow m_1 = 7$

  $k_2 = \{20, 30, 25\} \rightarrow m_2 = 25$

- **Step 5 : Reassign and find cluster mean.**

  $k_1 = \{2, 3, 4, 10, 11, 12\} \rightarrow m_1 = 7$

  $k_2 = \{20, 30, 25\} \rightarrow m_2 = 25$

- **Step 6 : Stop. The clusters in step 4 and 5 are same.**

  Final answer: $k_1 = \{2, 3, 4, 10, 11, 12\}$ and $k_2 = \{20, 30, 25\}$

Module 4

UEx. 4.4.2 MU - June 2021

Use K-means algorithm to create 3 - clusters for given set of values :{2, 3, 6, 8, 9, 12, 15, 18, 22}

☑ **Soln. :**

- **Step 1 : Randomly partition data into three clusters and calculate the mean value for each cluster.**

  $k_1 = \{2, 8, 15\} \rightarrow m_1 = 8.3$

  $k_2 = \{3, 9, 18\} \rightarrow m_2 = 10$

  $k_3 = \{6, 12, 22\} \rightarrow m_3 = 13.3$

- **Step 2 : Reassign the data points to each cluster and find again the cluster mean.**

  $k_1 = \{2, 3, 6, 8, 9\} \rightarrow m_1 = 5.6$

  $k_2 = \{\ \} \rightarrow m_2 = 0$

  $k_3 = \{12, 15, 18, 22\} \rightarrow m_3 = 16.75$

- **Step 3 : Reassign and find cluster mean**

  $k_1 = \{3, 6, 8, 9\} \rightarrow m_1 = 6.5$

  $k_2 = \{2\} \rightarrow m_2 = 2$

  $k_3 = \{12, 15, 18, 22\} \rightarrow m_3 = 16.75$

- **Step 4 : Reassign and find cluster mean**

  $k_1 = \{6, 8, 9\} \rightarrow m_1 = 7.6$

  $k_2 = \{2, 3\} \rightarrow m_2 = 2.5$

  $k_3 = \{12, 15, 18, 22\} \rightarrow m_3 = 16.75$

- **Step 5 : Reassign and find cluster mean**

  $k_1 = \{6, 8, 9\} \rightarrow m_1 = 7.6$

  $k_2 = \{2, 3\} \rightarrow m_2 = 2.5$

  $k_3 = \{12, 15, 18, 22\} \rightarrow m_3 = 16.75$

- **Step 6 : Stop. The clusters in step 4 and 5 are same.**

  Final answer: $k_1 = \{6, 8, 9\}$, $k_2 = \{2, 3\}$, $k_3 = \{12, 15, 18, 22\}$

16/10/25 * Answers:-

Q. Differentiate between Apriori Algorithm and FP-Growth Algorithm
→

Parameters	Apriori Algo.	FP-Growth Algo.
Storage structure	Array based	Tree based
Search Type	Breadth first search	Depth first search
No of database Scans	multiple scans (k+1) for generating candidate sets.	only twice scans for constructing frequent pattern tree.
Memory Utilization	Large Memory Space (candidate genⁿᵗⁿ)	Less memory space (No candidate gen)
Runtime	More time	Less time
Database	sparse/dense datasets	Large & medium datasets
Approach	Apriori utilizes a Level-wise approach, where it generates patterns containing 1 item, 2 items then 3 items & so on.	FP Growth utilizes a pattern-growth approach, it only considers patterns actually existing in db
Performance	Perform poorly on large datasets.	Works well on large datasets
Cost Computational	High	Low

Q. Explain Apriori Algorithm with eg flowchart.
→

- Apriori Algorithm is a popular algorithm used for association rule mining in data mining.
- Apriori Algorithm was first algorithm that was proposed for frequent itemset mining.
- It was later in 1994 improved by Rakesh Agrawal and Ramkrishnan Srikant and came to be known as Apriori.
- Apriori algorithm is a sequence of steps to be followed to find the most frequent itemset in the given database.
- This algorithm uses two steps "join" & "prune" to reduce the search space.

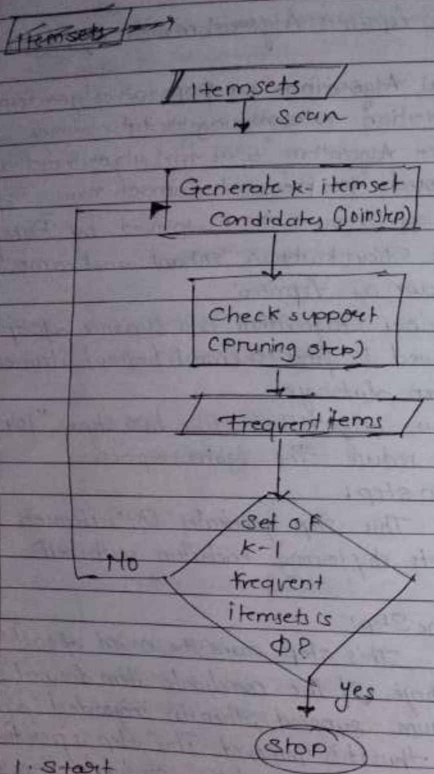
i) Join step:

This step generates (k+1) itemsets from k itemsets by joining each item with itself.

ii) Prune Step:

This step scans the count of each item in the database. If the candidate item does not meet minimum support, then it is regarded as Infrequent and thus it is removed. This step is performed to reduce the size of the candidate itemsets.

- This data mining technique follows the join and the prune steps iteratively until the most frequent itemset is achieved.



1. Start
2. Scan the datasets to generate candidate (itemsets).
3. Generate k-itemsets candidates (Joinstep):
The algorithm generates candidate itemsets of size k by joining frequent itemsets of size k-1
4. Check Support (Prunestep):
The algorithm checks the support of these candidate itemsets and prunes those that don't meet the minimum support threshold.
5. Frequent items: The remaining itemsets after pruning are considered frequent items.
6. Check if set of k-1 frequent itemsets is Φ?
   - If yes, the process stops.
   - If No, the algorithm continues to generate the next set of candidate itemsets.

Q] Given the foll. data, apply the apriori algorithm.
Given support threshold = 50%, confidence = 60%.

Transaction	List of items
T1	I1, I2, I3
T2	I2, I3, I4
T3	I4, I5
T4	I1, I2, I4
T5	I1, I2, I3, I5
T6	I1, I2, I3, I4

→ Support threshold = 50%.

∴ min-sup = 0.5 × no. of transaction
= 0.5 × 6
= 3

Thus, min_sup = 3

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Step 1: Count of Each Itemset (C_1) by scanning the database

Itemset	count	
{I_1}	4	
{I_2}	5	
{I_3}	4	
{I_4}	4	
{I_5}	2	<3

Step 2: Prune Step (L_1): C_1 shows that I_5 itemset does not meet min_sup=3, thus it's deleted only

Itemset	count
{I_1}	4
{I_2}	5
{I_3}	4
{I_4}	4

Step 3: Join Step: Form C_2 from $L_1 \bowtie L_1$

Itemset	Count	
{I_1, I_2}	4	
{I_1, I_3}	3	
{I_1, I_4}	2	<3
{I_2, I_3}	4	
{I_2, I_4}	3	
{I_3, I_4}	2	<3

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Step 4: Prune Step (L_2): C_2 shows that itemset (I_1, I_4) & (I_3, I_4) doesnot meet min_sup, thus it is discarded.

Itemset	Count
{I_1, I_2}	4
{I_1, I_3}	3
{I_2, I_3}	4
{I_2, I_4}	3

Step 5: Join step: Form C_3 from L_2 using $L_2 \bowtie L_2$

Itemset	Count.	
{I_1, I_2, I_3}	3	
{I_1, I_2, I_4}	2	<3
{I_1, I_3, I_4}	1	<3
{I_2, I_3, I_4}	2	<3

Step 6: Prune step (L_3): C_3 shows that itemsets {I_1, I_2, I_4} {I_1, I_3, I_4} & {I_2, I_3, I_4} doesnot meet min_sup, thus it is deleted.

Itemset	Count
{I_1, I_2, I_3}	3

Thus {I_1, I_2, I_3} is frequent.

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$I_3 \leftarrow \{I_1, I_2\}$
$I_2 \leftarrow \{I_1, I_3\}$
$I_1 \leftarrow \{I_2, I_3\}$

Step 3: Generate Association Rule:

- $\{I_1, I_2\} \rightarrow \{I_3\}$

$\therefore$ confidence $= \dfrac{\text{support}\{I_1, I_2, I_3\}}{\text{support}\{I_1, I_2\}} \times 100$

$= \dfrac{3}{4} \times 100$

$= 75\%$

- $\{I_1, I_3\} \rightarrow \{I_2\}$

$\therefore$ Confidence $= \dfrac{\text{support}\{I_1, I_2, I_3\}}{\text{support}\{I_1, I_3\}} \times 100$

$= \dfrac{3}{3} \times 100$

$= 100\%$

- $\{I_2, I_3\} \rightarrow \{I_1\}$ $= \dfrac{3}{4} \times 100 = 75\%$

- $\{I_1\} \rightarrow \{I_2, I_3\}$

$\therefore$ confidence $= \dfrac{\{I_1, I_2, I_3\}}{\{I_1\}} \times 100$

$= \dfrac{3}{4} \times 100 = 75\%$

- $\{I_2\} \rightarrow \{I_1, I_3\}$

$= \dfrac{3}{5} \times 100 = 60\%$

- $\{I_3\}$ $= \dfrac{3}{4} \times 100 = 75\%$

This shows that all the above association rules are strong if min. confidence threshold is 60%.

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Q) Explain FP-Growth Algorithm with its advantage and disadvantage

→ - FP Growth Algorithm stands for Frequent pattern growth algorithm.
- This algorithm is an improvement to the Apriori method.
- A frequent pattern is generated with the need for candidate generation.
- FP growth algorithm represents the database in the form of a tree called a frequent pattern tree or FP tree.
- This tree structure will maintain the association betn the itemsets.
- How FP Growth works:
  1. Build FP tree: This algo. scans the dB and builds FP tree, which is a compressed representn of the database.
  2. Mine FP-tree: Algo. recursively mines the FP tree to generate frequent itemsets.

• Advantages:-

i) Faster: pairing not done of items here, so this makes this algo. faster.

ii) It is efficient & scalable for mining both long & short frequent patterns.

iii) Disadvantages:-

i) It is expensive

ii) FP Growth more complex to implement than Apriori.

iii) Not suitable for incremental mining, where dB is updated frequently.

Q. Difference betn.

	Classification	Clustering.
i)	classificn is supervised learning	Clustering is unsupervised learning
ii)	In Classification algo. like decision trees, Bayesian classifiers are used.	In clustering algo. like, k-mean, k- method expectn: Maximization used.
iii)	Classification has prior knowledge of classes.	Cluster doesn't have any prior knowledge of classes.
iv)	eg:) Classificn betn gender.	eg:) discovery of patterns.
v)	Classificn requires training data to build the model.	Classification does not require training data.
vi)	Classificn is used in spam filtering, medical dignosis.	Clustering is used in Customer segmentation, market analysis, etc

4) Describe in detail Market Basket Analysis.

Ans:- 1) Market Basket Analysis (MBA) is a data mining technique used to find relationships or patterns between items that are frequently bought together by customers.

2) It helps businesses understand which products are often purchased together, so they can make better marketing, pricing and placement decisions.

3) It is also known as Association Rule Learning and it is commonly used in supermarkets, online shopping and retail stores.

4) The main goal of Market Basket Analysis is to study customers buying habits and find connections between different products.

For example:- If many customers who buy bread also buy butter, the store can place them close together or offer a combo discount.

Key Term in Market Basket Analysis:

1) Itemset :- A collection of one or more items ...gs
Example : {Milk, Bread}, {Soap, Sha...
2) Confidence : It shows how often items are bought together.
3) Support : It tells how often an item or itemset appears in all transactions.
4) Lift : It measures how strong the relationship is between items.

Applications of MBA:-

1) Retail and supermarkets.
2) E-commerce websites.
3) Banking
4) Telecommunication
5) Healthcare

Techniques of web structure mining:-

1) Analyzing Hyperlinks.
2) Graph Theory methods
3) Social Network Analysis
4) PageRank Algorithm
5) Link Analysis
6) Community detection

Applications:-

1) Search Engine Optimization (SEO)
2) website Navigation Improvement
3) Recommendation System
4) Fraud Detection

Define cluster Analysis. Give the application of cluster analysis.

Cluster Analysis is a method used in statistics and data analysis to group objects, people or data items in such a way that the object in the same group (called a cluster) are more similar to each other than to those in other groups. It is an unsupervised learning technique used to find natural patterns or structures in data.

Application of Cluster Analysis:-

1) Customer Segmentation :- Grouping customers based on purchasing behavior for targeted marketing.
2) Image Processing :- Grouping similar pixels for image segmentation.
3) Document classification :- Grouping documents with similar topics or content.
4) Medical Diagnosis :- Grouping patients with similar symptoms or disease patterns.
5) Social Media :- It can group people with similar interest or behaviors - for example, users who like the same kind of posts.
6) Education :- schools can group students based on their learning styles or performance to give better support.

Cluster Analysis

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1) AdBhay Stransactions. Let minimum support = 2 count and minimum confidence = 60%. find all frequent items using Apriori Algorithm. Also list strong association rules.

Mayes

TID	Items
100	1,3,4
200	2,3,5
300	1,2,3,5
400	2,5
500	1,3,5

2) Explain kmeans Algorithm with diagram. Use k-means algorithm to create 3-clusters for given set of value:

Mayes

{2,3,6,8,9,12,15,18,22}

3) consider 4 objects with 2 attribute (X & Y). These 4 objects are to be grouped together into 2 clusters using k-means clustering algorithm. Foll. are objects with their Attribute values.

Decy

Object	X	Y
A	1	1
B	2	1
C	4	3
D	5	4

4) Given training data for Height Classification. Classify the tuple t<Rohit, M, 1.95> using Naive Bayes Classification

Decy

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Name	Gender	Height	Output
Kiran	F	1.6m	Short
Jatin	M	2m	Tall
Madhuri	F	1.9	Med
Manisha	F	1.88m	Med
Shilpa	F	1.7m	Short
Bobby	M	1.85	Med
Kavita	F	1.6m	short
Dinesh	M	1.7m	Short
Rahul	M	2.2m	Tall
Shree	M	2.1m	Tall
Divya	F	1.8m	Med
Tushar	M	1.95m	Med
Kim	F	1.9m	Med
Aarti	F	1.8m	Med
Rajashree	F	1.75m	Med

* Rohit M 1.95

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Naive Bayes.

(P1) Rohitsum => Rohit, M, 1.95, ?

Class labelled is: Height Classification.

C_1 = Short = 4

C_2 = Med = 8 Total = 15.

C_3 = Tall = 3.

$\therefore P(C_1) = \frac{4}{15}$. $\therefore P(C_2) = \frac{8}{15}$ $\therefore P(C_3) = \frac{3}{15}$

Let Event X_1 be Gender = 'M'

X_2 be Height = 1.95

* Compute $P(X|C_1)$

Using Naive Bayesian classification

$\therefore P(X|C_1) = P(X_1|C_1)\ P(X_2|C_1)$

$\therefore P(X_1|C_1) = P\left(\frac{\text{Gender}=M}{\text{Short}}\right) = \frac{1}{4}$

$\therefore P(X_2|C_1) = P\left(\frac{\text{Height}=1.95}{\text{Short}}\right) = \frac{0}{4}$

$\therefore P(X|C_1) = \frac{1}{4} \times \frac{0}{4} = 0$

$\therefore P(X|C_1) \times P(C_1) = 0 \times \frac{4}{15} = 0$

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* Compute $P(X|C_2)$

Using NB classificatn.

$\therefore P(X|C_2) = P(X_1|C_2)\ P(X_2|C_2)$

$\therefore P(X_1|C_2) = P\left(\frac{\text{Gender}=M}{\text{Med}}\right) = \frac{2}{8}$

$\therefore P(X_2|C_2) = P\left(\frac{\text{Height}=1.95}{\text{Med}}\right) = \frac{1}{8}$

$\therefore P(X|C_2) = \frac{2}{8} \times \frac{1}{8} = 0.031$ or $\frac{1}{32}$

$\therefore P(X|C_2) \times P(C_2) = \frac{1}{32} \times \frac{8}{15} = \frac{1}{60} = 0.01667$

\# Compute $P(X|C_3)$

Using Naive Bayes Classification

$\therefore P(X|C_3) = P(X_1|C_3)\ P(X_2|C_3)$

$\therefore P(X_1|C_3) = P\left(\frac{\text{Gender}=M}{\text{Tall}}\right) = \frac{3}{3}$

$\therefore P(X_2|C_3) = P\left(\frac{\text{Height}=1.95}{\text{Tall}}\right) = \frac{1}{3}$ (1.9, 2.0)

$\therefore \frac{3}{3} \times \frac{1}{3} = P(X|C_3) = \frac{1}{3}$

$\therefore P(X|C_3)\ P(C_3) = \frac{1}{3} \times \frac{3}{15} = \frac{1}{15} = 0.0667$

So tall $0.0667 > 0.0167 > 0$.

Naive Bayesian Classification

Apply Statistical based algorithm to obtain the actual probabilities of each event to classify new tuple as 'Buy.computer = Yes'. Use the given data:

Classify X = (Age = '<=30', Income = 'Medium', Student = 'yes', Credit Rating = 'fair').

Id	Age	Income	Student	Credit. Rating	Buy.computer
1	<=30	High	No	Fair	No
2	<=30	High	No	Excellent	NO
3	31..40	High	No	Fair	Yes
4	>40	Medium	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
6	>40	Low	Yes	Excellent	NO
7	31..40	Low	Yes	Excellent	Yes
8	<=30	Medium	No	Fair	NO
9	<=30	Low	Yes	Fair	Yes
10	>40	Medium	Yes	Fair	Yes
11	<=30	Medium	Yes	Excellent	Yes
12	31..40	Medium	No	Excellent	Yes
13	31..40	High	Yes	Fair	Yes
14	>40	Medium	No	Excellent	No
*	<=30	Medium	Yes	Fair	Yes

Class label attribute is Buy.computer

C1 : Buys computer = Yes : 9 Samples

C2 : Buys computer = No : 5 Samples

$\therefore P(C_1) = \frac{9}{14}$ & $P(C_2) = \frac{5}{14}$

Let Event X_1 be Age = '<=30'

———— X_2 - Income = Medium

———— X_3 - Student - Yes &

———— X_4 - Credit Rating = Fair

* Compute $P(X|C_1)$ using Naive bayesian Classification

$P(X|C_1) = P(X_1|C_1)P(X_2|C_1)P(X_3|C_1)P(X_4|C_1)$

$P(X_1|C_1) = P\left(\frac{Age<=30}{Buy\text{-}comp=Yes}\right) = \frac{2}{9}$

$P(X_2|C_1) = P\left(\frac{Income=Medium}{Buy\text{-}comp=Yes}\right) = \frac{4}{9}$

$P(X_3|C_1) = P\left(\frac{Student=Yes}{Buy\text{-}comp=Yes}\right) = 6/9$

$P(X_4|C_1) = P\left(\frac{Credit\ Rating=Yes}{Buy\text{-}comp=Yes}\right) = 6/9$

$P(X|C_1) = \frac{2}{9} \times \frac{4}{9} \times 6/9 \times 6/9 = 0.044$

$P(X|C_1)P(C_1) = 0.044 \times 9/14 = \underline{0.028}$

* Compute $P(X|C_2)$ using Naive bayesian Classification

$P(X|C_2) = P(X_1|C_2)P(X_2|C_2)P(X_3|C_2)P(X_4|C_2)$

$P(X_1|C_2) = P\left(\frac{Age<=30}{Buy\text{-}comp=No}\right) = \frac{3}{5}$

$P(X_2|C_2) = P\left(\frac{Income=Medium}{Buy\text{-}comp=No}\right) = \frac{2}{5}$

$P(X_3|C_2) = P\left(\frac{Student=Yes}{Buy\,comp=No}\right) = \frac{1}{5}$

$P(X_4|C_2) = P\left(\frac{Credit\text{-}Rating=Fair}{Buy\text{-}comp=No}\right) = \frac{2}{5}$

$P(X|C_2) = \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{2}{5} = 0.019$

$P(X|C_2)P(C_2) = 0.019 \times 5/14 = \underline{\underline{0.007}}$

$P(X|C_1)P(C_1) > P(X|C_2)P(C_2)$

$\therefore$ X = (Age = '<=30', Income = 'Medium', Student = 'Yes', Credit Rating = 'Fair') belongs to class Buys.computer = Yes

Apriori Algo.

Q—

TID	Items
100	1,3,4
200	2,3,5
300	1,2,3,5
400	2,5
500	1,3,5.

min support = 2 ; confidence = 60%.

→ Step 1:- Counting each itemset (C1) we get

Itemset	Count	
1	3	
2	3	
3	4	
4	1	<2
5	4.	

Step 2:- Prune step.

Itemset	count
1	3
2	3
3	4
5	4

~~Step 3:- Join step.~~

~~Itemset~~

Step 3: Joinset

item set	count	
1,2	1	<2
1,3	3	
1,5	2	=2
2,3	2	=2
2,5	3	
3,5	3	

Step 4: Prune step.

Itemset	count.
1,3	3
1,5	2
2,3	2
2,5	3
3,5	3

1,2,3,5

Step 5:- Joinstep.

Itemset	count.	
1,2,3	1	< 2
1,2,5	1	< 2
1,3,5	2	
2,3,5	1	< 2

Step 6: Prune step

Itemset	count
1,3,5	2.

Step 7 :- Generate Association Rule

1, 3, 5

1 ↔ 1, 3, 5

3 ↔ 1, 5

5 ↔ 1, 3

- $\{I1, I3\} \rightarrow \{I5\}$

$$\text{confidence} = \frac{1,3,5}{1,3} \times 100$$

$$= \frac{2}{3} \times 100$$

$$= 67\%$$

strong.

- $\{I3\} \rightarrow \{I1, I5\}$

$$= \frac{1,3,5}{13} \times 100$$

$$= \frac{2}{4} \times 100$$

$$= 50\%$$